Auditable Memory and Financial Identification Limits in Electronic Limit Order Books

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Abstract

When trading algorithms condition on recent prices, price formation is a closed-loop system. We ask what an outside observer can learn about the algorithms from market data, and at what cost. In a linear cohort model, prices identify only a merged rate–weight measure. Separating two memory rates δ apart costs executed quantity of order δ^{-2} and trading capital of order δ^{-4}, with explicit constants; separating K unknown cohorts costs $(\Delta\eta)^{-(4K-2)}$ experiments, a rate attained by Prony's method; with general completely monotone memory the minimax accuracy is only of order $1/\log N$. Class-tagged disclosure removes the barrier and reduces the sample requirement by a factor of order $(\Delta\eta)^{-(4K-4)}$. In data, trader-class positions published by Binance and, in a

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pre-registered test, by the CFTC for eight regulated futures markets pin down memories that prices leave unrestricted.

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Reproducibility. Every number in the paper is produced by a script or notebook cell that writes a result file from public data or from a closed form; Appendix D maps each table, figure and number to its source. Every simulation number was re-run for this version.

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1 Introduction

Regulators increasingly require firms to say which algorithm sent each order. Under MiFID II, EU investment firms must flag algorithmic orders and identify the algorithm responsible for each investment and execution decision in their transaction reports (ESMA, 2016); the US Consolidated Audit Trail records who stands behind every order (SEC, 2012). These regimes are costly to build and to comply with, and they rest on an implicit claim: that a supervisor could not learn the same thing from public prices and order flow. This paper tests that claim, and prices the answer.

The setting is a closed-loop market. Algorithms update on recent prices, their orders move prices, and prices feed back into the algorithms. We model K cohorts of algorithms that track the price with exponential memory, the reduced form of steady-state Kalman filtering and of regularized online gradient descent (Propositions 2.1–2.2), and ask what an outside observer can learn about them from market data. The answer is sharp. Prices reveal only a merged summary of the algorithms. Separating two algorithms whose memories differ by δ costs trading capital of order δ^{-4}, and separating K unknown algorithms costs a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$; with general memory, accuracy improves only as $1/\log N$, so halving the error squares the number of experiments. Class-tagged disclosure removes the barrier, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$. In data, Binance's published long/short ratios by trader class, a coarse form of class tagging, pin down memories that prices and flow leave unrestricted in each of 30 futures markets, and the CFTC's mandated trader-class reports do the same in eight regulated U.S. futures markets.

1. An identification gap. Prices identify only the merged rate–weight measure $M = \sum_i q_i \eta_i \, \delta_{1-\eta_i}$ of the algorithms (Theorem 3.1). Cohorts that share a memory rate are observationally equivalent, whatever their identities or wealth. Nearby rates can be separated, but at a cost that rises quickly as they approach each other.

2. A resource gap. For a known symmetric pair separated by δ, price-only identification with deterministic probes requires executed quantity of order δ^{-2}, and a single pulse attains this order (Theorem 4.1). Because a probe moves the price against the prober, the quantity order becomes a capital order δ^{-4}, whether capital is supplied by a counterparty under a margin contract (Theorem 4.3) or by the probe trader itself (Theorem 4.5). We then compute what an account actually needs. A spectral argument gives a lower bound on the trading loss of any flat round trip (Theorem 5.2); Wald's identity extends it to adaptive, sequential probing (Theorem 5.3); and a Rayleigh–Hölder dual bounds the finite-horizon optimum (Theorem 5.4). At the calibrated parameters the bracket is $[61.41, 64.48]$ par units. For K unknown cohorts the number of independent experiments is exactly of order $(\Delta\eta)^{-(4K-2)}$: a two-point lower bound (Theorem 6.3) is matched by Prony's method for every K (Theorem 6.4). For general completely monotone memory the minimax accuracy is of exact order $1/\log N$: a lower bound built from moment-matched cohort pairs (Lemma 6.5) is matched by a moment estimator solved as a linear program (Theorem 6.6), so the exponential-memory model is the optimistic case.

3. A measurement gap. Passive data do not escape the barrier. The closed-loop transfer function is minimum phase, so the price spectrum identifies the merged measure (Theorem 6.7), and natural excitation can separate a pair with zero controlled quantity (Proposition 4.6). But the information rate falls at the same polynomial

order as for active probing (Theorem 6.8). Only richer disclosure removes the dependence on rate spacing: tagged flow (Proposition 7.1) or audited equity (Proposition 4.4). Tagging reduces the sample requirement by a factor of order $(\Delta\eta)^{-(4K-4)}$ for rates (Theorem 7.2). On Binance, published long/short ratios by trader class pin down class memories 28 to 124 times more precisely than prices and flow in each of 30 futures markets (Section 7.2), and calibrating the theorem to the BTCUSDT class memories makes them worth about 333 times as much data as prices alone.

4. What feedback indices measure. Standard reduced-form feedback indices are dominated by filtering mechanics and clock alignment; a placebo pipeline reproduces the mechanical persistence exactly; measured feedback rises with market depth across 30 perpetuals; and two fee events that multiply trading volume by 2.6 and divide it by 4.9 leave measured feedback unchanged (Section 9).

5. A spectral blind spot. Even without identification, could a supervisor monitor risk through eigenvalues or volatility? No. Markets with identical eigenvalues, one-step volatility and stationary price variance can have arbitrarily large worst-case transient loss (Theorem 8.1). This motivates the open question with which the paper ends: what can be measured and controlled without identifying the algorithms? The companion paper takes it up.

What we do not claim. The identification results hold for the stated linear model and observation contracts. They are not universal limits on surveillance with participant-level records, which can be far more informative (Propositions 4.4 and 7.1). The matching upper bound holds on a class with separated rescaled rates and weights bounded away from zero. Capital figures depend on the observation-noise calibration, which we estimate rather than assume. The empirical results are

reduced-form diagnostics; they do not identify structural learning parameters. Our pre-registered real-data test of the identification boundary failed its criterion on one of its three parts: on 42 days of data the merged response was precisely estimated in only 5 of 30 markets (Section 7.2). Binance's class ratios are positions, not tagged order flow, so they are the closest public analogue to the disclosure of Proposition 7.1, not an instance of it.

1.1 Related literature

Price formation and impact. Our price equation is a discrete linear impact model in the tradition of Kyle (1985), Glosten and Milgrom (1985) and Hasbrouck (1991), in which order flow moves prices and prices revert toward value. Transient impact and the cost of trading follow Almgren and Chriss (2001), Obizhaeva and Wang (2013), Gatheral (2010) and Bouchaud et al. (2004); Alfonsi et al. (2016) show that absence of price manipulation requires positive-definite decay kernels, which is the property Lemma 4.2 delivers in our cohort economy. Bouchaud et al. (2018) document order splitting and long memory in order flow, which is the main confounder in Section 9. Relative to this literature we treat the impact kernel not as a primitive but as the closed-loop output of adaptive traders, and ask what it reveals about them.

Algorithmic trading and feedback. Algorithms that react to recent prices can amplify or dampen moves. Brunnermeier and Pedersen (2009) and Shleifer and Vishny (1992) study liquidity spirals and fire sales, and Kirilenko et al. (2017) document the behaviour of automated intermediaries in the 2010 Flash Crash. Sornette et al. (2023) show that non-normal interactions can generate transient bubbles. Our contribution is on the measurement side: we show what a closed-loop observer can and cannot learn about the feedback, and that standard reduced-form indices are

dominated by filtering and timing.

Closest work. Three literatures are closest. Super-resolution and moment-based mixture estimation give rates for recovering clustered point sources and mixing measures from noisy moments (Batenkov et al., 2021; Liu and Ammari, 2022; Wu and Yang, 2020; Kong and Valiant, 2017). Our K-cohort rate is, in that language, the square of the amplitude condition number $(\Delta\eta)^{-(2K-1)}$ for K clustered nodes, and we do not claim the exponent as new. What is new is the following: the observable is a closed-loop price response, whose zeros rather than poles carry the memory rates, so the moment problem must first be extracted from the loop (Theorem 3.1); the matching lower and upper bounds are proved for real nodes in the power-moment problem with explicit constructions (Gauss and Gauss–Radau pairs, Prony's method); the general-memory rate is shown to be logarithmic, with both bounds; and the rates are mapped into executed quantity, trading capital and the value of disclosure, which is what a supervisor needs. The microstructure literature infers trading intentions from public prices and flow (Kyle, 1985; Glosten and Milgrom, 1985; Hasbrouck, 1991); we show what such inference cannot recover about adaptive algorithms and at what cost. The regulatory literature on audit trails motivates disclosure; we supply a quantitative value for it.

Statistical identification. The identification results use two-point testing arguments (Le Cam, 1986; Tsybakov, 2009). The moment-matching construction of Section 6 is the dynamical analogue of the slow rates in finite-mixture estimation (Manole and Ho, 2022) and of super-resolution of near-colliding sources (Liu and Ammari, 2022; Batenkov et al., 2021); the general-memory result combines best-approximation duality (Bernstein, 1914; Rivlin, 1981) with the moment comparisons used by Kong

and Valiant (2017) and Wu and Yang (2020), and its logarithmic rate parallels super-smooth deconvolution (Fan, 1991); recovery of exponentials connects to ESPRIT and matrix-pencil methods (Roy and Kailath, 1989; Hua and Sarkar, 1990). Minimum-phase spectral factorization and the Whittle likelihood are classical (Whittle, 1953). What is new here is the mapping from these statistical rates into executed quantity and money through a financially specified account, and the result that the relevant rates are zeros, not poles, of the observable transfer function.

Non-normal dynamics and supervision. Transient growth of stable non-normal systems is classical (Trefethen and Embree, 2005; Kreiss, 1962). We use it to show that supervision through eigenvalues and second moments of prices is blind to transient risk (Theorem 8.1), which motivates the identification-free control of the companion paper. On the regulatory side, pre-trade risk controls for broker-dealers with market access are governed by SEC Rule 15c3-5 (SEC, 2010), and participant-level audit trails are the natural disclosure regime for Proposition 7.1. A large literature studies how pre- and post-trade transparency affects liquidity and trading costs (Pagano and Röell, 1996; Boehmer et al., 2005; Bessembinder et al., 2006); we ask a different question, what disclosure lets a supervisor learn about the algorithms that trade, and give the answer in samples. Crypto venues, whose market structure is studied by Makarov and Schoar (2020), are a natural laboratory for this question because they publish class-level positioning every five minutes, while the CFTC publishes it weekly for regulated futures.

Partial fractions, Gaussian testing, convolution inequalities and positive-system gains are established tools. Our contribution is to connect them into a quantitative chain from memory resolution, through probing resources, to money, with matching orders, explicit constants and a supervisory interpretation, and then to measure the

Table 1: Map of results

Question	Answer	Where
What do prices identify?	The merged rate–weight measure only	Thm 3.1
Quantity to separate a known pair	$\Theta(\delta^{-2})$, exact error formula	Thm 4.1
Capital to separate a known pair	$\Theta(\delta^{-4})$, counterparty or own funds	Thms 4.3, 4.5
Constant at the calibration	$[61.41, 64.48]$ par units; $\approx 10^5$ USDT	Thms 5.2–5.4, Table 6
Does adaptivity help?	At most by a factor ≈ 0.49 in expected loss	Thm 5.3
Unknown mixtures	$(\Delta\eta)^{-(4K-2)}$, tight for every K	Thms 6.3, 6.4
General memory	Accuracy $\asymp 1/\log N$, lower and upper bounds	Lem. 6.5, Thm 6.6
Passive data	Identifies M; same polynomial cost	Thms 6.7, 6.8, Prop. 4.6
What removes the barrier?	Tagged flow, audited equity	Props 4.4, 7.1
What is tagging worth?	Factor $(\Delta\eta)^{-(4K-4)}$; $\approx 333\times$ for BTCUSDT	Thm 7.2
Do class data pin down memories?	Yes: 28–124× tighter in 30 of 30 markets	Table 15
Can eigenvalues certify safety?	No	Thm 8.1
What do feedback indices measure?	Mostly filtering and timing; small real feedback	Tables 18, 20
Does feedback track activity?	No: fees move volume 2.6–4.9×, not $\hat{Q}$	Table 22

relevant quantities in data.

1.2 Outline

Section 2 sets out the model. Section 3 gives the identification boundary. Section 4 links memory resolution to quantity and money for a known pair. Section 5 computes account-level capital. Section 6 treats unknown mixtures and passive data. Section 7 studies disclosure. Section 8 gives the spectral blind spot. Section 9 reports the empirical evidence. Section 11 concludes. Proofs are in Appendix A; Appendix C collects additional results, and Appendix D maps every number to its source.

2 Model and microfoundations

Time is discrete. Let p_t be the price, p^* the fundamental value and $x_t = p_t - p^*$ the pricing error. Competitive dealers absorb net order flow with price impact $\kappa > 0$,

and arbitrage pulls the error toward zero with persistence $a \in [0, 1)$:

$$x_t = ax_{t-1} + \kappa \sum_{i=1}^{K} q_i y_{i,t-1} + \kappa v_t + w_t, \tag{1}$$

where v_t is an executed probe quantity and w_t a price innovation. Cohort i holds a state $y_{i,t}$ with exponential memory and trades $q_i y_{i,t}$:

$$y_{i,t} = (1 - \eta_i)\, y_{i,t-1} + \eta_i x_t, \qquad \eta_i \in (0, 1). \tag{2}$$

The ordering is *price first*: the price responds to the probe, then cohorts update. Write $r_i = 1 - \eta_i$ for the memory rate and $w_i = q_i \eta_i$ for the merged weight. The feedback kernel and its transform are

$$k_m = \sum_i w_i r_i^m, \qquad G(z) = \sum_i \frac{w_i z}{z - r_i}, \tag{3}$$

and the transfer function from probes to prices is

$$H(z) = \frac{\kappa}{z - a - \kappa G(z)}, \tag{4}$$

with impulse response $h_0 = \kappa, h_1, h_2, \ldots$. Stacking $z_t = (x_t, y_{1,t}, \ldots, y_{K,t})^\top$ gives $z_t = J z_{t-1} + b v_t$ with

$$J = \begin{pmatrix} a & \kappa q_1 & \cdots & \kappa q_K \\ \eta_1 a & 1 - \eta_1 + \eta_1 \kappa q_1 & \cdots & \eta_1 \kappa q_K \\ \vdots & & \ddots & \vdots \\ \eta_K a & \eta_K \kappa q_1 & \cdots & 1 - \eta_K + \eta_K \kappa q_K \end{pmatrix}, \qquad S \equiv a + \kappa \sum_i q_i. \tag{5}$$

In the *positive economy* ($q_i \geq 0$), J is entrywise nonnegative, its price row sums to S, and the economy is Schur stable iff $S < 1$.

Proposition 2.1 (Kalman memory)**.** *A desk that tracks a latent random-walk trend* $v_{t+1} = v_t + \zeta_{t+1}$, $\zeta \sim N(0, \sigma_\zeta^2)$, *from signals* $r_t = v_t + \nu_t$, $\nu \sim N(0, \sigma_\nu^2)$, *using the steady-state Kalman filter, updates as* $\hat{v}_{t+1} = (1-\eta)\hat{v}_t + \eta r_t$ *with*

$$\eta = \frac{-\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}}{2\sigma_\nu^2} \in (0, 1). \tag{6}$$

Proposition 2.2 (Gradient memory)**.** *Online gradient descent with* L_2 *regularization* $\gamma \in (0, 1)$, *step size* $\alpha > 0$ *and loss* $\frac{1}{2}H(x_t - \theta_t)^2$ *gives* $\theta_{t+1} = (1-\eta)\theta_t + \eta q x_t$ *with* $\eta = \gamma + \alpha H \in (0, 1)$ *and* $q = \alpha H/\eta$. *Tabular Q-learning with learning rate* η *has the same form.*

Remark 2.3 (Why exponential memory). Propositions 2.1–2.2 show that two different design philosophies, Bayesian filtering and regularized online learning, produce the same observable reduced form. The memory rate encodes the desk's signal-to-noise ratio or regularization, which is exactly the private information a supervisor would like to recover. Section 3 shows why prices reveal it only in merged form.

Lemma 2.4 (Positive price-response geometry)**.** *In the positive, stable economy the impulse response is a nonnegative mixture of decaying exponentials,*

$$h_j = \sum_\ell \omega_\ell \lambda_\ell^j, \qquad \omega_\ell \geq 0, \quad \sum_\ell \omega_\ell = \kappa, \quad 0 \leq \lambda_\ell < 1.$$

It is completely monotone and nonincreasing; its peak is the direct impact κ, *even though* J *is non-normal. In price/memory coordinates the price-only gain is* $\sup_t \|e_x^\top J^t\|_2 = 1$.

Numerically, $(-1)^n \Delta^n h_j \geq 0$ for $n \leq 4$ held in all 892 stable random positive economies tested. Lemma 2.4 matters twice: it makes probing costs positive (Section 4), and it shows that in this economy transient amplification of the price response cannot arise from cohort feedback alone. The companion paper locates amplification in order-book depth instead.

2.1 Discussion of the assumptions

Linearity. The model is linear in prices, flows and memory states. Linearity is what makes the merged-measure equivalence exact and the testing problems Gaussian. Nonlinear strategies (for example threshold rules or position limits) can break the equivalence class, but they do so in regions of the state space that a small probe does not visit; for small probes the linearization around the steady state is the relevant object.

Price-first timing. We order the price update before the cohort update. Under the alternative lagged timing the transition matrix becomes $\left(\begin{smallmatrix} a & 1 \\ \eta Q & 1-\eta \end{smallmatrix}\right)$ and its eigenvalues differ, so stability statements must be computed under the timing actually used. The identification results depend on the transfer function and survive either convention.

Exponential memory. Propositions 2.1–2.2 justify exponential memory for filtering and regularized learning. Neural policies are not exponential filters; how closely their local linearizations follow exponential memory is an empirical question we do not settle here. Mixtures of exponentials can approximate any completely monotone kernel, so a power-law memory corresponds to a continuum of cohorts, for which Theorem 6.6 shows that accuracy improves only logarithmically in the data.

Algorithms that react to being probed. The probing results assume that the cohorts keep their memories while they are probed. An algorithm that detects a probe could pause, shorten its memory or trade against the probe. Four observations limit what this does to our conclusions. First, a reaction that all cohorts share does not separate them, and an algorithm that hides by mimicking another rate makes the hypotheses closer, so neither lowers the costs we derive; a reaction that differs across cohorts is itself a signature, but one the observer must model, which is a new identification problem rather than a free one. Second, the passive route needs no probe: Theorem 6.8 shows that natural price variation identifies the same separation at the same polynomial order in sample length, so an observer who fears detection pays in time, not in rate. Third, detection is itself a testing problem against the same noise: the generalized-eigenvector probe of Section 5 trades at most 1.25 par units in any period, and a probe sized within normal flow variation is hard to tell from noise, so the threat bites mainly for large single pulses, which our ledger shows are not optimal anyway. Fourth, what we cannot rule out is an algorithm designed to defeat surveillance, which turns identification into a game between observer and observed; our results are the benchmark against which such a game would be measured, and we leave it to future work.

Known primitives. The resource theorems take κ, a and the noise law as known. Unknown primitives can only make separation harder, so the lower bounds remain valid, while the upper bounds require estimating them, which is cheap relative to the memory rates because they enter at first order.

Observation contracts. Each result names what the observer sees. A price-only observer, an observer with executed quantities, one with noisy statements, one with

audited equity and one with tagged flow face different experiments. The paper's central message is that these differences, not the observer's statistical skill, determine what can be learned.

3 The identification boundary

A central question in algorithmic market surveillance is whether an observer of executions and prices can identify the individual strategies active in the market.

Theorem 3.1 (Merged rate–weight equivalence)**.** *Let $\kappa > 0$ and $a \in [0, 1)$ be known. Use zero initial states and a common probe contract, and observe the modeled price state x_t.*

(i) Equivalence class. *Two economies $\mathcal{E}^A = \{(q_i^A, \eta_i^A)\}$ and $\mathcal{E}^B = \{(q_j^B, \eta_j^B)\}$ produce identical price responses for every probe sequence if and only if their merged measures coincide:*

$$\sum_{i:\eta_i^A=\eta} q_i^A \eta \;=\; \sum_{j:\eta_j^B=\eta} q_j^B \eta \qquad \forall \eta \in (0, 1). \tag{7}$$

The minimal realization has dimension $1 + |\mathrm{supp}(M)|$, counting distinct rates with nonzero merged weight after signed cancellation. A cohort with $\eta = 1$ merges into the anchor a without adding a pole.

(ii) Separation complexity under average power. *Consider testing a single cohort with rate $\bar{\eta}$ against two cohorts with rates $\bar{\eta} \pm \delta$ and the same total weight, on a compact family that is uniformly Schur stable with rates bounded away from 0 and 1. Observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma_\nu^2)$ and use deterministic probes*

with $\frac{1}{T}\sum_{t<T} v_t^2 \le P$. *Any test with worst-case error at most* $\alpha \in (0, 1/4)$ *needs*

$$T \;\ge\; \frac{4\sigma_\nu^2(1-2\alpha)^2}{K^2 P\,\delta^4} = \Omega(\delta^{-4}), \tag{8}$$

where K *depends only on the family.*

The memory rates are poles of G and *zeros* of H, not closed-loop poles. This has a practical consequence we verify in Section 6: pole-fitting methods applied directly to the price impulse response recover closed-loop eigenvalues (0.4256 and 0.8224 for $K = 1$, $\eta = 0.3$, $a = 0.5$, $\kappa = 0.8$, $q = 0.2$) and miss the memory rate 0.70.

Remark 3.2 (Scope). The result concerns the stated model and observation contract. Cohorts sharing an exact rate enter prices through their merged weight; this does not determine identities or wealth allocations. Nearby rates can be distinguishable with enough information. The δ^{-4} statement in (ii) is a lower bound on a declared family, with family-dependent constants. Individual algorithms may choose their rates strategically, but any unilateral change maps into the same observer-facing merged measure, which reinforces rather than circumvents the boundary.

4 Probing resources, memory resolution and money

Observation length alone cannot buy identification when the lifetime quantity of controlled executions is fixed. This section connects the merged-memory experiment to a financially specified input/output contract. Throughout, Φ is the standard normal distribution function.

4.1 Quantity

Theorem 4.1 (Sharp probing-resource orders for a known symmetric pair)**.** *Fix* $0 \le a < 1$, $\kappa, q > 0$, $S = a + \kappa q < 1$ *and* $\eta \in (0,1)$. *Compare a single cohort of weight* q *and rate* η *with two cohorts of weight* $q/2$ *and rates* $\eta \pm \delta$, *both in* $[\epsilon, 1-\epsilon]$. *All parameters are known. The price receives executed quantity through* κv_t, *starts from zero, has no process noise, and is observed with independent* $N(0,\sigma^2)$ *noise. Probes are deterministic and common to both alternatives. Let* $d_T = (h_j^\delta - h_j^0)_{j<T}$ *and* $K_* = 2\kappa^2 q/[\epsilon^3(1-S)^2]$.

(i) For every probe with $\sum_t |v_t| \le V$,

$$D_{\mathrm{KL}}(P_0^v \| P_\delta^v) \le \frac{K_*^2 V^2 \delta^4}{2\sigma^2}, \qquad e_{\mathrm{worst}} \ge \frac{1}{2}\Big(1 - \frac{K_* V \delta^2}{2\sigma}\Big)_+. \tag{9}$$

(ii) The optimal worst-case error over these probes and all classifiers is exactly

$$e_T^*(V) = \Phi\left(-\frac{V\|d_T\|_2}{2\sigma}\right), \tag{10}$$

attained by a pulse at the first input and the likelihood-ratio test. Waiting adds information but $\|d_T\|_2 \uparrow \|d_\infty\|_2 < \infty$, *so it cannot make the error arbitrarily small at fixed* V. *Error at most* $\alpha < 1/2$ *requires* $V \ge 2\sigma(1-2\alpha)/(K_*\delta^2)$ *independently of* T.

(iii) The first two taps coincide and $d_2 = -\kappa^2 q\delta^2$. *A single pulse of size*

$$A = \frac{2\sigma\Phi^{-1}(1-\alpha)}{\kappa^2 q\delta^2} \tag{11}$$

and the third observation achieve error exactly α. *The necessary and sufficient quantity order is* δ^{-2}.

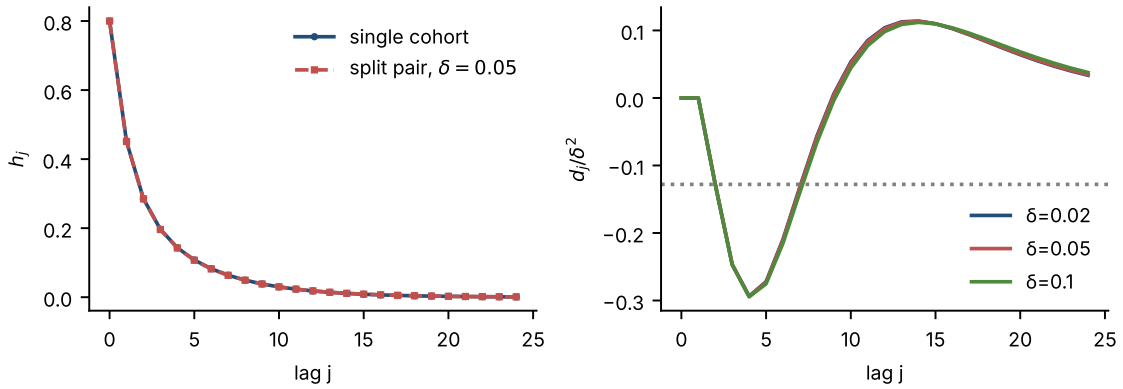


Figure 1: Left: impulse responses of the single cohort and the split pair ($\delta = 0.05$) are nearly indistinguishable. Right: the difference scaled by δ^2 collapses onto one curve, with first nonzero value $-\kappa^2 q$ at lag 2 (dotted). Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

(iv) Under average power $\sum_{t<T} v_t^2 \le PT$*, the necessary horizon order is* δ^{-4}*, attained by the pulse whenever* $T \ge \max\{3, \lceil A^2/P \rceil\}$.

(v) For a fixed monetary price exposure $\lambda \neq 0$*, additive reserve* L *and cost envelope* $c\sum|v_t|$*, the sharp robust quantity allowance under budget* $B > L$ *is* $V_B = (B - L)/(|\lambda|\kappa + c)$.

The identity in (iii) is verified to machine precision for $\delta \in \{0.02, 0.05, 0.10\}$: the difference is -1.2×10^{-4}, -7.5×10^{-4} and -3.0×10^{-3} at $\kappa = 1$, $q = 0.3$, and -3.2×10^{-4} at the calibration $\kappa = 0.8$, $q = 0.2$, $\delta = 0.05$. The theorem does not solve estimation of an unknown mixture (Section 6).

A Monte Carlo check of the exact error. Formula (10) is exact for the likelihood-ratio test. We simulated 200,000 noisy paths per hypothesis at the calibration ($\sigma = 0.01$, $\delta = 0.05$) for several pulse sizes and horizons (Table 2); simulated errors match the formula to Monte Carlo precision under both hypotheses.

Figure 1 shows why the problem is hard. The two economies have the same direct

Table 2: Exact known-pair error: formula versus simulation

Pulse V	Horizon T	$\Phi(-V\|d_T\|_2/2\sigma)$	Simulated error, H_0	Simulated error, H_δ
20	3	0.3745	0.3734	0.3762
40	3	0.2611	0.2599	0.2603
40	10	0.0029	0.0030	0.0027
80	20	1.8×10^{-10}	0	0

200,000 paths per hypothesis; $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$.

impact and the same first-lag response; they first differ at lag two, by an amount proportional to δ^2, and the whole difference sequence scales with δ^2. A test therefore needs the signal-to-noise ratio $V\|d\|/\sigma$ to grow like δ^{-2}, which is the quantity order, and because the adverse mark of a probe grows with the square of its size, capital grows like δ^{-4}.

4.2 Execution cost

A probe changes inventory as well as price. We therefore specify cash and execution timing. Price has units of currency per asset, and κ currency per squared quantity.

Lemma 4.2 (Fair execution)**.** *For a quote schedule refreshed after the uncontrolled state update, execution at the average price of a linear book costs, relative to the constant fundamental mark,*

$$C(v) = \tfrac{1}{2} v^\top T_h v + c\|v\|_1, \qquad (T_h)_{ij} = h_{|i-j|}. \tag{12}$$

In the positive stable economy T_h is positive definite, so no nonzero round trip produces a deterministic gain.

We verify the ledger identity (12) against an explicit cash account to 5.7×10^{-13} on 1,000 random probes, and the smallest eigenvalue of T_h over 1,000 random stable positive economies is 0.141. A direct-only proxy $\sum \frac{1}{2}\kappa v_t^2$ omits cross-time impact: for $v = (+1, -1)$ it gives 0.80 against a true Toeplitz loss of $h_0 - h_1 = 0.3488$.

4.3 Capital supplied by a counterparty

Theorem 4.3 (Sharp robust quantity cap from cash and margin)**.** *Use the positive, stable, reset economy with constant fundamental price $P_* > 0$ and no process noise. A designated dealer fills only nonnegative probes $v_t \geq 0$ at the average price $P^0_{t+1} + \kappa v_t/2$ of a linear schedule, holds inventory $I_{t+1} = I_t - v_t$ and cash $K_{t+1} = K_t + v_t(P^0_{t+1} + \kappa v_t/2) - cv_t$, starts with equity $B > 0$, and must satisfy $\mu|I_t|P_t \leq E_t$ at every time. For the robust contract admitting every probe with $\sum_t v_t \leq V$, the largest allowance guaranteeing margin is*

$$V_B = \frac{2B}{\beta + \sqrt{\beta^2 + 4(\mu + \frac{1}{2})\kappa B}}, \qquad \beta = c + \mu P_*, \tag{13}$$

equivalently $(\mu + \frac{1}{2})\kappa V^2 + (c + \mu P_*)V \leq B$. The cap holds across all positive cohort representations with $S < 1$, without resolving individual rates. An observer that receives only noisy prices and executed quantities has best error $\Phi[-V_B\|d_T\|_2/(2\sigma)]$ within this allowance; fixed error $\alpha < 1/2$ requires capital of order at least δ^{-4}, and a single positive pulse attains that order.

Even with zero initial inventory, the probe creates a quadratic adverse mark that a fixed initial price exposure would miss. Figure 2 illustrates the experiment.

Proposition 4.4 (Exact financial disclosure changes the experiment)**.** *In this noiseless-state account, exact disclosure of the third equity mark after a single positive pulse $0 < A \leq V_B$ distinguishes the known pair with zero error for every $\delta > 0$.*

Exact marked equity is extra information, not a free observation assumed by the price-only theorem. Real account marks have their own precision: with reports rounded to 0.01, a guaranteed change requires a larger trade (Section 7).

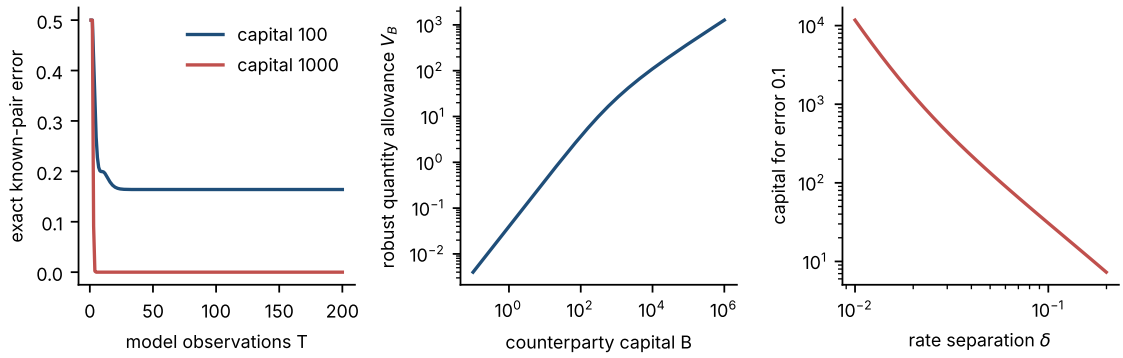


Figure 2: Known-pair experiment funded by the counterparty cash/margin contract. (a) Exact Gaussian error at fixed capital 100 and 1,000, separation $\delta = 0.05$. (b) The sharp robust quantity allowance (13). (c) Capital for error 0.1 with 200 observations. Parameters: $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, observation s.d. 0.003, reference price 100, margin 0.25, cost 0.02. No process noise.

4.4 Capital paid by the probe trader

Theorem 4.5 (Matching capital order for funded deterministic probes)**.** *Use the same economy and execution law. The statistical observer receives only noisy prices and executed quantities. The probe trader starts with cash $B > 0$ and zero inventory; a finite signed sequence v changes inventory by $+v_t$ and cash by $K_{t+1} = K_t - v_t(P^0_{t+1} + \kappa v_t/2) - c|v_t|$, after which trading stops and the trader must remain solvent as the price returns to P_*. Put*

$$r = 1 - \epsilon(1 - S) < 1, \qquad m = \kappa \frac{1 - r}{1 + r} > 0, \qquad K_* = \frac{2\kappa^2 q}{\epsilon^3 (1 - S)^2}.$$

Every solvent strategy satisfies $D_{\mathrm{KL}}(P^v_0 \| P^v_\delta) \le K_*^2 B \delta^4/(\sigma^2 m)$, so error at most $\alpha < 1/2$ requires

$$B \;\ge\; \frac{2\sigma^2 m (1 - 2\alpha)^2}{K_*^2 \delta^4}, \tag{14}$$

without a quantity cap and at any horizon. Conversely, for any margin fraction $0 < \mu \le 1$, the pulse A in (11) achieves error α and remains margin-solvent at every

later mark with capital $B_A = \frac{\kappa}{2}A^2 + (c + \mu P_*)A$. *Necessary and sufficient capital orders are* δ^{-4}.

Theorem 4.5 is an order statement: an elementary time-domain coercivity argument for Toeplitz forms that establishes the δ^{-4} rate before the exact frequency-domain optimization of Section 5. Its explicit lower constant is correspondingly loose: at the calibrated parameters it evaluates to 4.9×10^{-4} par units at $\delta = 0.05$ with $\epsilon = 0.35$, five orders of magnitude below any design we can execute. Section 5 computes the constant that matters.

4.5 A worked example: two Kalman desks

To make the orders concrete, consider two desks that track the same trend with Kalman filters (Proposition 2.1) and differ only in their signal-to-noise ratios: $\sigma_\zeta^2/\sigma_\nu^2 = 0.09$ and 0.16. Their gains are $\eta_1 = 0.2584$ and $\eta_2 = 0.3279$, so $\bar{\eta} = 0.2931$ and $\delta = 0.0348$. With equal weights, the economy is observationally the split pair of Theorem 4.1 around $\bar{\eta}$, and the response difference first appears at lag two, $d_2 = -\kappa^2 q \delta^2 = -1.55 \times 10^{-4}$ (verified).

A supervisor who wants to tell this market from a single desk with gain $\bar{\eta}$ at 5% error using a single pulse and the third observation needs $A = 2\sigma\Phi^{-1}(0.95)/(\kappa^2 q\delta^2) = 212$ units of executed quantity, and, under a 0.5% margin and 4 basis points of fees, single-pulse capital of 18,166 par units. The pulse is very inefficient: by the δ^{-4} scaling of Theorem 5.2, the optimal flat probe needs about $61.41 \times (0.05/0.0348)^4 \approx 262$ par units. Converting with the measured Roll noise (factor $(0.40/0.01)^2$), the optimal probe costs roughly 4×10^5 USDT and the naive pulse about 3×10^7 USDT. Two desks whose signal-to-noise ratios differ by almost a factor of two are therefore expensive, but not impossible, to tell apart by probing; three or more desks are much harder

(Theorem 6.3).

4.6 Natural excitation and the cost of ambiguity

Proposition 4.6 (Passive distinction with zero controlled quantity)**.** *Keep the known pair, but add independent uniform innovations $w_t \in [-b, b]$ of variance $s_w^2 = b^2/3$ at each price update, observe $\tilde{x}_t = x_t + \nu_t$ with $\nu_t \sim N(0, \sigma^2)$, use independent three-step reset blocks and no probes, with $P_* > 3b$. Let $b_1 = a + \kappa q\eta$, $b_{2,0} = b_1^2 + \kappa q\eta(1-\eta)$ and $b_{2,\delta} = b_{2,0} - \kappa q\delta^2$. In each block the covariance of the first and third observations differs by $\Delta = \kappa q s_w^2 \delta^2$. With*

$$R = \max_{r \in \{0,\delta\}} \Big\{ (s_w^2 + \sigma^2)\big[s_w^2(1 + b_1^2 + b_{2,r}^2) + \sigma^2\big] + s_w^4 b_{2,r}^2 \Big\},$$

the midpoint test of the average product has worst-case error at most α whenever $n \geq 4R/(\alpha\Delta^2)$ blocks are observed: zero controlled quantity and order δ^{-4} blocks.

The Chebyshev bound is conservative but the order is real. With $b = 0.03$, observation s.d. 0.01 and $\delta = 0.10$, the bound asks for 3.9×10^7 blocks at $\alpha = 0.1$; in simulation the midpoint test still errs 32–38% of the time with 2×10^5 blocks. Independent resets and known innovation laws are extra conditions. The proposition does not show that anonymous public records identify participants; it shows that passive excitation can accumulate memory information while the probe account stays untouched.

Proposition 4.7 (Conditional quote and allocation cost of ambiguity)**.** *Consider a one-unit ticket with adverse mark $Z_\theta \sim N(\mu_\theta, \sigma_\theta^2)$ and a CARA dealer with risk aversion $\gamma > 0$. Let $c_\theta = \mu_\theta + \gamma\sigma_\theta^2/2 \geq 0$ with $\sup_\theta c_\theta < \infty$. Under competition with zero robust certainty equivalent, the informed quote is $s_1 = c_{\theta_0}$ and the ambiguous*

quote $s_2 = \sup_\theta c_\theta$*, so* $\Delta = s_2 - s_1 \geq 0$*, strictly when the worst relevant cost exceeds* c_{θ_0}*. Under inverse demand* $v(V) = (V_0 - V)/b$ *and marginal cost* c*, raising an interior quote from* s_0 *to* $s_0 + \Delta$ *lowers surplus by* $W_{\text{loss}} = b(s_0 - c)\Delta + \frac{b}{2}\Delta^2$*.*

Cohort non-identification alone does not establish that the worst relevant cost exceeds c_{θ_0}; the set of relevant costs is an additional primitive. Treating a risk charge as a social cost requires a defended welfare criterion.

5 Account-level capital

Theorem 4.5 gives the order but not the constant. This section computes what a trader actually needs under a calibrated linear-impact environment, then converts it to Bitcoin dollars.

5.1 Observation design and the tri-region partition

An observer distinguishes two candidate mechanisms with rates θ_0 and $\theta_1 = \theta_0 + \delta$ over N periods, submitting probes $V = (V_1, \ldots, V_N)^\top$ that generate inventory $I_t = \sum_{s \leq t} V_s$. Under contract I_0 (active probing, Gaussian observation noise) $p_t = (h_i * V)_t + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, so

$$D_{\text{KL}}(P_0^{\otimes N} \| P_1^{\otimes N}) = \frac{1}{2\sigma^2} \sum_{t=1}^{N} \big((h_1 * V)_t - (h_0 * V)_t\big)^2. \tag{15}$$

The observer minimizes capital $C(N, V) = \sum_t (\lambda V_t^2 + \phi|V_t|) + \max_t \mu p_0 |I_t|$ subject to (i) separation $D_{\text{KL}} \geq \frac{1}{2}(z_\alpha + z_\beta)^2$, (ii) caps $|V_t| \leq V_{\text{eff}}$ and $|I_t| \leq I_{\max}$, and (iii) a solvency budget $\max_t \text{Loss}_t(V) \leq B$. Here $V_{\text{eff}} = \min(V_{\max}, D_{\max}/\kappa)$ combines resting depth and a price-displacement limit.

Proposition 5.1 (Tri-region partition and finite-session resolution floor)**.** *Under observation horizon T per pulse, (δ, B) space splits into three regions.*

(i) Feasible single pulse. *For $\delta \geq \delta_{\min}(T)$ and $B \geq C^*(\delta)$ one pulse within capacity suffices. For a simple rate shift ($\eta_1 = \eta_0 + \delta$) the response difference is $\Theta(\delta)$ and $C^* = \Theta(\delta^{-2})$; for symmetric splitting the first-order variation cancels, the difference is $\Theta(\delta^2)$ and $C^* = \Theta(\delta^{-4})$.*

(ii) Sequential bounded probing. *For $\delta < \delta_{\min}(T)$ the required quantity exceeds V_{eff}. Capped pulses each supply divergence $V_{\text{eff}}^2 \|d_T\|^2/(2\sigma^2)$, so*

$$N_{\text{pulses}}(T, \delta) = \left\lceil \frac{4\sigma^2 z_\alpha^2}{V_{\text{eff}}^2 \|d_T\|_2^2} \right\rceil. \tag{16}$$

(iii) Infeasible with isolated pulses. *If the budget cannot finance sequential costs, peak inventory breaches $I_{\max}$, or a session deadline binds, the finite-session floor $\delta_{\min}(T_{\text{session}})$ is the root of $V_{\text{eff}} \|d_{T_{\text{cycle}}}(\delta)\|_2 = 2\sigma z_\alpha / \sqrt{\lfloor T_{\text{session}}/T_{\text{cycle}} \rfloor}$, with small-$\delta$ approximation $\delta_{\min} = \Theta(T_{\text{session}}^{-1/4})$.*

At the calibration $\kappa = 0.8$, $q = 0.2$, $\sigma = 0.01$, $V_{\max} = 50$, $D_{\max} = 5$ ($V_{\text{eff}} = 6.25$), the single-tap floor is $\delta_{\min}(3) = \sqrt{2\sigma z_\alpha/(V_{\text{eff}}\kappa^2 q)} = 0.2028$; extended observation lowers it to 0.1376 ($T = 4$), 0.0973 ($T = 10$), 0.0914 ($T = 20$) and 0.0905 ($T = 100$). At $\delta = 0.05$ the simplified one-leg benchmark (cost $\frac{1}{2}\kappa V_{\text{eff}}^2 + \phi P_* V_{\text{eff}} = 15.875$ per pulse plus initial margin 3.125) needs 271 pulses and 4,305.25 par units with $T = 3$, 58 pulses and 923.88 with $T = 4$, but only 12 pulses and 193.63 with $T = 20$ and 11 pulses and 177.75 with $T = 100$; optimizing the pulse size (6.06 rather than 6.25) lowers the $T = 20$ figure to 182.03. Most of the saving comes from observing the transient response rather than a single tap (Figure 3).

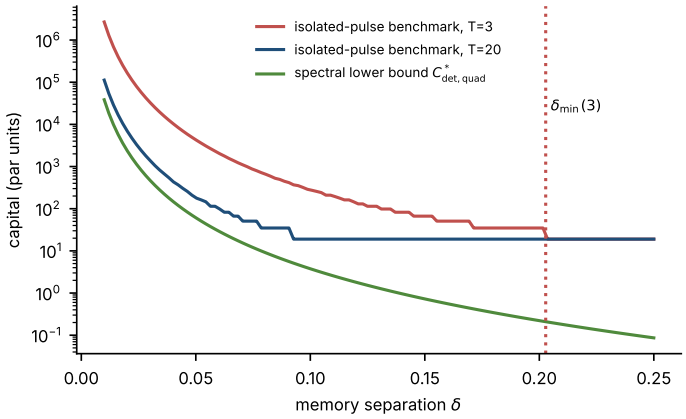


Figure 3: Capital against separation: isolated-pulse benchmarks with three and twenty observations per pulse, and the spectral lower bound of Theorem 5.2. The dotted line is the single-tap resolution floor $\delta_{\min}(3) = 0.2028$. Recomputed in this revision.

These are costs within an isolated-pulse design class, not minimum capital; the next subsection replaces the formula by a ledger.

5.2 Account-level capital and the gap to the lower bound

We run each candidate design through an explicit synthetic ledger at reference price $P_* = 100$: fills at the average linear-book price, taker fees of 4 basis points, marks at the prevailing price, and 0.5% maintenance margin. Initial capital is $C_0 = \max_t(\mathrm{margin}_t - \mathrm{equity}_t)$, maximized over both hypotheses, and the test uses the full price path including unwind legs.

Table 3 gives three conclusions. First, twelve buys that are never unwound fail the 5% target once the test uses the joint price path (divergence $4.95 < 5.41$) and end with 75 units of inventory. Second, if every unwind leg is itself informative, five or six round trips meet the target with zero terminal inventory. Third, pulses are not optimal: a generalized-eigenvector probe, which maximizes divergence per unit of fee-free round-trip cost, needs about 65 par units with maximum trade 1.25 and

Table 3: Account-level initial capital at $\delta = 0.05$ (par units)

Design	C_0	Error	End inv.	Peak margin	Max $\lvert v_t \rvert$	Trades
12 buys, $T = 20$ apart, never unwound	229.13	0.0578	75.00	39.38	6.25	12
6 buy/sell round trips	189.45	0.0308	0.00	3.28	6.25	12
5 buy/sell round trips	157.89	0.0442	0.00	3.28	6.25	10
12 buys, each unwound after 20 steps	378.82	0.0040	0.00	3.28	6.25	24
Generalized-eigenvector probe ($L = 120$)	65.06	0.0500	0.00	2.74	1.24	121
Generalized-eigenvector probe ($L = 160$)	64.86	0.0500	0.00	2.17	1.09	161
Fee-refined probe ($L = 240$)	64.48	0.0500	0.00	—	1.30	241

The pulse designs place a pulse every 40 steps and close it 20 steps later; the fee-refined probe is re-optimized by sequential quadratic programming over $L = 240$. The certified lower bound of Theorem 5.2 is 61.41; the explicit constant of Theorem 4.5 is 4.9×10^{-4}.

maximum displacement 2.23, well inside limits.

For flat fee-paying policies $v \in \mathbb{R}^L$ with $\sum_t v_t = 0$, the capital problem is

$$\min_{v} \max_{i \in \{0,1\}} L_i(v) + \sum_t \phi P_* |v_t| + \max_t \mu P_* |I_t| \quad \text{s.t.} \quad \|(H_1 - H_0)v\|_2 \geq 2\sigma z_{1-\alpha}, \quad (17)$$

where $L_i(v) = \frac{1}{2} v^\top T_{h_i} v$ is the exact Toeplitz round-trip loss. Zero-padding embeds horizon L into $L + 1$ at identical cost and information, so the minimum is non-increasing in L.

Theorem 5.2 (Spectral dissipation lower bound)**.** *Let $\hat{H}_i(z) = \sum_{j \geq 0} h_{i,j} z^{-j}$, $\Delta\hat{H} = \hat{H}_1 - \hat{H}_0$, and define the dissipation kernel*

$$g_i(\omega) = \operatorname{Re} \hat{H}_i(e^{i\omega}) - \kappa/2 > 0 \qquad \forall \omega \in [-\pi, \pi]. \quad (18)$$

(i) Toeplitz energy identity. *For any flat round trip, $L_i(v) = \frac{1}{2\pi} \int_{-\pi}^{\pi} g_i(\omega) |V(e^{i\omega})|^2 d\omega$.*

(ii) Lower bound. $\|(H_1 - H_0)v\|_2^2 \leq \sup_\omega \frac{|\Delta\hat{H}(e^{i\omega})|^2}{g_i(\omega)} L_i(v)$, *so every deterministic*

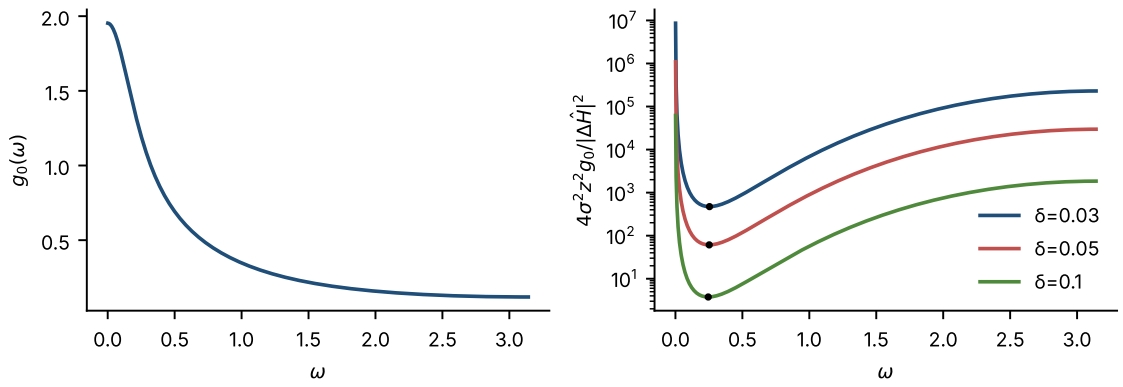


Figure 4: Left: the dissipation kernel $g_0(\omega)$ is strictly positive. Right: the capital ratio in (19); dots mark its minimum, near $\omega^* \approx 0.25$ for every separation. Recomputed in this revision.

round trip with error at most α has

$$L_i(v) \;\geq\; 4\sigma^2 z_{1-\alpha}^2 \inf_{\omega} \frac{g_i(\omega)}{|\Delta\hat{H}(e^{i\omega})|^2} \;\equiv\; C^*_{\mathrm{det,quad}}(\delta). \tag{19}$$

(iii) Bracket. *At $\delta = 0.05$, $C^*_{\mathrm{det,quad}} = 61.14$ at $\omega^* = 0.2526$; adding the unavoidable fee $\phi P_* \cdot 2\sigma z / \sup_\omega |\Delta\hat{H}|$ gives the certified bound $C_{\mathrm{det}} \geq 61.41$. A feasible 240-step sequence attains 64.48, so the fee-paying optimum lies in* $[61.41, 64.48]$.

Positivity of g_i follows from Lemma 2.4: $g_i(\omega) = \frac{1}{2}\sum_\ell \omega_\ell \frac{1-\lambda_\ell^2}{1-2\lambda_\ell \cos\omega + \lambda_\ell^2} > 0$. We compute $C^*_{\mathrm{det,quad}}$ across separations (Table 4): 61.14 at $\delta = 0.05$, with $\delta^4 C^*_{\mathrm{det,quad}}$ between 3.67×10^{-4} and 3.84×10^{-4}, confirming the δ^{-4} law. The fee term, the feasible upper bounds and the adaptive bound of Theorem 5.3 are reported in the same table.

Theorem 5.3 (Adaptive and sequential probing)**.** *Let $(v_t)_{t\leq\tau}$ be any non-anticipating policy adapted to the price filtration, with an almost surely finite stopping time and*

Table 4: Capital lower bounds and feasible upper bounds across separations (par units)

δ	ω^*	$C^*_{\mathrm{det,quad}}$	$\delta^4 C^*_{\mathrm{det,quad}}$	Certified C_{det}	Feasible UB	Adaptive LB
0.02	0.2546	2397.85	3.84×10^{-4}	2399.58	2428.25	1176.04
0.03	0.2541	473.19	3.83×10^{-4}	473.97	484.50	232.51
0.05	0.2526	61.14	3.82×10^{-4}	61.41	64.48	30.22
0.08	0.2488	9.26	3.79×10^{-4}	9.37	10.38	4.64
0.10	0.2453	3.77	3.77×10^{-4}	3.83	4.44	1.91
0.15	0.2331	0.73	3.67×10^{-4}	0.76	1.01	0.38

Calibration $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.4$, $\sigma = 0.01$, $\alpha = 0.05$. From `recompute_theory.py` and `recompute_p1_capital.py`; the certified column adds the fee lower bound to the fee-free infimum; the feasible column is the best flat probe found over $L \leq 240$; the adaptive column adds the fee lower bound to the adaptive bound.

$\sum_{t \leq \tau} v_t = 0$. *Then*

$$\mathbb{E}_0[L_0(v)] \;\geq\; 2\sigma^2 \,\mathrm{kl}(\alpha, 1-\beta) \inf_{\omega} \frac{g_0(\omega)}{|\Delta \hat{H}(e^{i\omega})|^2}, \tag{20}$$

where $\mathrm{kl}(\alpha, 1-\beta) = \alpha \log \frac{\alpha}{1-\beta} + (1-\alpha) \log \frac{1-\alpha}{\beta}$. *Hence adaptation can lower expected fee-free loss by at most the factor* $\mathrm{kl}(\alpha, 1-\beta)/(2z^2_{1-\alpha}) \approx 0.4897$ *at* $\alpha = \beta = 0.05$*, and cannot alter the* δ^{-4} *order.*

At $\delta = 0.05$ the adaptive bound is 29.94, or 30.22 with the minimal fee. Pairing it with the feasible deterministic 64.48 gives a certified ratio of at least 0.464. In 20,000 simulated sequential probability-ratio tests, which split the optimal probe into ten copies of equal information, mean loss was 39.5 under H_0, between the two bounds, but losses vary with the stopping time: the 95th and 99th percentiles of required account capital were 91.7 and 128.4, and funding a trader with only the mean loss gives a 37.5% insolvency rate before the test ends. Expected-loss savings are not solvency savings.

Theorem 5.4 (Finite-horizon Rayleigh–Hölder duality)**.** *For horizon* $L \geq 2$*, let* E *be the zero-sum first-difference basis spanning flat probes,* $Q_i = \frac{1}{2} T_{h_i}$ *(so that*

Table 5: Finite-horizon capital at $\delta = 0.05$ (par units)

Horizon L	Lower bound $C_{\mathrm{quad}}(L)$	Best feasible probe	Gap to $C_{\mathrm{det}} = 61.41$
80	62.55	65.02	5.9%
120	61.84	64.59	5.2%
160	61.54	64.50	5.0%
240	61.32	64.48	5.0%
320	61.25	≤ 64.48	$\leq 5.0\%$
480	61.19	≤ 64.48	$\leq 5.0\%$

Quadratic bound $c_{\mathrm{req}}/\lambda_L$ from the generalized eigenproblem of Theorem 5.4 on flat probes (equivalently $c_{\mathrm{req}}/(2\lambda'_L)$ with λ' computed against T_{h_0} rather than $\frac{1}{2}T_{h_0}$); it converges to the infinite-horizon value 61.14. Feasible probes: top generalized eigenvector refined by sequential quadratic programming on the full ledger; at $L = 320$ and 480 the $L = 240$ probe padded with zeros is feasible at the same cost, so the best feasible value can only be lower.

$L_i(v) = v^\top Q_i v$) and $A_{\mathrm{diff}} = (H_1 - H_0)^\top (H_1 - H_0)$, and let $G_i = E^\top Q_i E \succ 0$ and $M = E^\top A_{\mathrm{diff}} E \succeq 0$ be the reduced execution-loss and information matrices, with G_i-orthonormal generalized eigenpairs $0 \leq \lambda_1 \leq \cdots \leq \lambda_L$, $c_{\mathrm{req}} = 4\sigma^2 z^2_{1-\alpha}$ and $c_{\mathrm{fee}} = \phi P_*$. Every feasible flat probe decomposes into its top-two spectral component and an orthogonal residual, and

$$L_i(v) + c_{\mathrm{fee}}\|v\|_1 \;\geq\; \frac{c_{\mathrm{req}}}{\lambda_L} + \max\!\left(0,\, F_{\mathrm{dom}} - \frac{W_\perp}{4(1-\gamma F_{\mathrm{dom}})}\right) \;\equiv\; C_{\mathrm{RH}}(L), \tag{21}$$

where F_{dom}, $W_\perp$ and γ are explicit functions of the top-two eigenpairs and the residual spectrum (Appendix A).

The bound is computable for any horizon. Table 5 reports its quadratic part, which decays toward the infinite-horizon value 61.14 at rate $O(L^{-1})$, and the best feasible fee-paying probe we find at each horizon; the gap between them is about 5% from $L = 160$ on. The fee of unpadded full-support eigenvectors grows as $\Theta(\sqrt{L})$, which is why zero-padded or L_1-localized probes are needed at long horizons.

5.3 Robustness: heteroskedastic impact and probing protocols

Corollary 5.5 (Invariance under impact-induced heteroskedasticity)**.** *Suppose probing inflates innovation variance as* $\sigma_{i,t}^2(v) = \sigma^2(1 + \xi[(h_i * v)_t]^2)$ *and slippage scales as* $\kappa_{\text{eff}} = \kappa(1 + \xi_0|v_t|)$*. For capacity-bounded probes,* $D_{\text{KL}}(P_0^v \| P_1^v) \le \frac{1+O(\xi^2 V_{\text{eff}}^2)}{2\sigma^2} \|(H_1 - H_0)v\|_2^2 = O(\|v\|_2^2 \delta^4)$*, and convex impact only raises dissipation. Required capital remains* $\Theta(\delta^{-4})$*.*

Four probing protocols (unconstrained single pulse, repeated capped pulses, a sequential test with dynamic throttling, and a predetermined generalized-eigenvector trajectory) all show capital slopes near -4 in $\log \delta$: -3.90, -3.84, -3.47 and -3.81 over $\delta \in [0.03, 0.12]$, where the sequential slope is flattened by the 1.7 pulses needed at $\delta = 0.12$ (Appendix Table 24). At equal observation horizon, sequential stopping saves 14–49% of capital relative to repeated capped pulses, but most of the apparent saving from the single-tap baseline comes from observing the transient response, not from adaptivity.

5.4 From par units to Bitcoin dollars

Capital scales as $\sigma^2 \delta^{-4}$, so the dollar magnitude hinges on the observation noise. Treating the 4 USDT ten-second price volatility as observation noise would give 9.8 million USDT at $\delta = 0.05$, but that conflates price volatility with measurement error. We now estimate observation noise from Binance BTCUSDT one-second closes on January 10–20, 2023, with the Roll covariance estimator $\hat{\sigma} = \sqrt{-\text{cov}(\Delta p_t, \Delta p_{t-1})}$ (Roll, 1984). Daily estimates range from 0.09 to 0.77 USDT, median 0.40 USDT.

At realistic noise, separating rates that differ by 0.05 costs about 9.9×10^4 USDT, which is not prohibitive for a well-capitalized desk (Table 6). The barrier becomes

Table 6: Capital lower bound under alternative observation-noise calibrations (USDT)

Noise calibration	$\delta = 0.02$	$\delta = 0.05$	$\delta = 0.10$
Roll estimator, median (0.40 USDT)	3,874,696	99,193	6,200
One-second price-change s.d., median (1.98 USDT)	93,688,369	2,398,422	149,902
Earlier assumption (4 USDT, upper scenario)	383,812,500	9,825,600	614,100

$C(\sigma, \delta) = 61.41 \times (\sigma/0.01)^2 \times (0.05/\delta)^4$, with σ the unrounded median. Noise estimated on 11 days of checksum-verified Binance BTCUSDT one-second klines.

Table 7: Daily observation-noise estimates, Binance BTCUSDT one-second closes

Date	Roll $\hat{\sigma}$ (USDT)	One-second change s.d. (USDT)	Mean price (USDT)
2023-01-10	0.152	0.679	17,288
2023-01-11	0.092	0.891	17,456
2023-01-12	0.232	2.292	18,343
2023-01-13	0.348	1.762	19,075
2023-01-14	0.773	3.511	20,818
2023-01-15	0.482	1.790	20,778
2023-01-16	0.629	2.357	21,036
2023-01-17	0.457	1.976	21,194
2023-01-18	0.695	2.541	21,122
2023-01-19	0.402	1.316	20,850
2023-01-20	0.316	2.038	21,306
Median	0.402	1.976	

The Roll estimator measures the negative first-order autocovariance of price changes, which reflects bid-ask bounce and recording noise rather than price volatility.

large only for finer separation (3.9 million USDT at $\delta = 0.02$) and, through Theorem 6.3, when several cohorts are unknown. Two factors we do not model would raise these figures: walking several levels of a real book costs more than the linear top-of-book impact assumed here, and impact-induced heteroskedasticity raises the noise a probe must overcome. The capital figures are therefore lower bounds for the stated model.

6 Unknown mixtures and passive data

Theorem 4.1 concerns a *known* pair. A supervisor faces an unknown number of cohorts with unknown rates and weights. This section asks how fast the cost of recovering

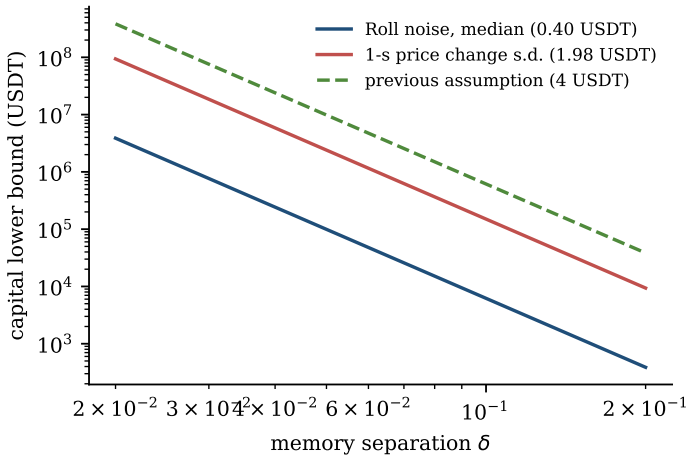


Figure 5: Capital lower bound against memory separation for three observation-noise calibrations.

them grows.

6.1 Why naive pole fitting fails

Because the memory rates are zeros of H (Theorem 3.1), extracting poles directly from the price impulse response by matrix-pencil or Prony methods targets closed-loop eigenvalues. A four-step inversion repairs this: (1) decouple $g_t = h_{t+1} - ah_t$; (2) form a Hankel matrix of g and compute a minimal realization by Kung's eigensystem realization; (3) compute its transmission zeros, which are the roots of the denominator of G; (4) project reconstructed flow onto the recovered memory filters to obtain the weights (Appendix C.2). Without noise this recovers $K = 0$, $K = 1$ and well-separated two-cohort systems exactly (Appendix C.2); with noise, per-cohort weights become unreliable once rates are close. The reason is information-theoretic, as the rest of this section shows.

Table 8: Cramér–Rao sample sizes for multi-cohort weight recovery ($T = 120$ taps)

Scenario	K	N^* probes	σ_{crit}	ML success ($f = 1$)	RMSE/CRB
$\eta = 0.30$	1	0.1	6.05×10^{-3}	94%	1.02
$\Delta\eta = 0.30$	2	47.8	2.31×10^{-4}	94%	1.01
$\Delta\eta = 0.20$	2	597	6.55×10^{-5}	96%	0.98
$\Delta\eta = 0.10$ ($\delta = 0.05$ pair)	2	40,374	7.96×10^{-6}	93%	1.03
$\Delta\eta = 0.05$	2	2,620,320	9.88×10^{-7}	95%	1.03
spacing 0.10	3	17,349,791	3.84×10^{-7}	98%	1.22
spacing 0.20	3	27,038	9.73×10^{-6}	98%	0.88

Tolerance ± 0.01 at 95%. σ_{crit} is the single-probe noise at which one probe suffices; operational single-probe noise is 1.6×10^{-3}. ML columns: least squares over all weights and rates from six generic starts (never the true values), 200 replications per row at σ_{crit}; success means every weight within ± 0.01. Fitted log–log slope of N^* in $\Delta\eta$: -6.04.

6.2 Local information: the Cramér–Rao benchmark

Proposition 6.1 (Cramér–Rao benchmark)**.** *Let θ collect the weights $w_1, \ldots, w_K$ and rates $\eta_1, \ldots, \eta_K$. Suppose the impulse response $h_t(\theta)$, $t \le T$, is observed with i.i.d. Gaussian error from N reset probes of size V_{probe} against innovation noise σ_p, so the error has standard deviation $\sigma_h = \sigma_p/(V_{\text{probe}}\sqrt{N})$. With Jacobian $J_{tj} = \partial h_t/\partial\theta_j$ and $F_{\text{unit}} = J^\top J$, any regular unbiased estimator has $\mathrm{Var}(\hat{w}_k) \ge \sigma_h^2 [F_{\text{unit}}^{-1}]_{kk}$, and a 95% interval of half-width* tol *requires*

$$N^* \ge \Big(\frac{1.96}{\text{tol}}\Big)^2 \Big(\frac{\sigma_p}{V_{\text{probe}}}\Big)^2 [F_{\text{unit}}^{-1}]_{kk}. \tag{22}$$

For $K = 2$ with spacing $\Delta\eta$, $[F_{\text{unit}}^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$, so $N^* = \Theta((\Delta\eta)^{-6})$.

At the critical noise level maximum likelihood attains the bound: its RMSE for the hardest weight is between 0.88 and 1.22 times the Cramér–Rao value and every weight is within tolerance in 93% to 98% of replications (Table 8), so the critical noise levels, not estimator failure, are what make close cohorts expensive. Under ordinary surveillance budgets ($N \le 50$) unbiased estimators cannot resolve close weights.

6.3 A global two-point bound for any estimator

Proposition 6.1 applies to regular unbiased estimators. Could biased, ridge, shrinkage or Bayesian estimators escape the $(\Delta\eta)^{-6}$ scaling? No.

Theorem 6.2 (Global two-point bound and matching upper bound, $K = 2$)**.** *Fix total weight $q = 0.20$, mean rate $\bar{\eta} = 0.40$, target weight separation $\tau = 0.01$, and $\Delta\eta \in (0, 8/11)$. Let $p_{\mp} = \frac{1}{2} \mp \tau/q$ and*

$$\theta_0 = (qp_-, qp_+, \bar{\eta} - p_+\Delta\eta, \bar{\eta} + p_-\Delta\eta), \qquad \theta_1 = (qp_+, qp_-, \bar{\eta} - p_-\Delta\eta, \bar{\eta} + p_+\Delta\eta). \tag{23}$$

(i) Both have spacing $\Delta\eta$, weight separation $\|w(\theta_1) - w(\theta_0)\|_\infty = 2\tau$, and equal rate moments $M_m = \sum_k w_k \eta_k^m$ for $m \in \{0, 1, 2\}$, while

$$M_3(\theta_1) - M_3(\theta_0) = 4\tau p_- p_+ (\Delta\eta)^3. \tag{24}$$

Hence $\|h(\theta_1) - h(\theta_0)\|_2 = \Theta(\tau(\Delta\eta)^3)$.

(ii) For any measurable estimator (including biased, ridge, James–Stein or MAP),

$$\sup_\theta P_\theta(\|\hat{w} - w(\theta)\|_\infty \geq \tau) \;\geq\; 1 - \Phi\Big(\frac{\sqrt{N}\|h(\theta_1) - h(\theta_0)\|_2}{2\sigma_h}\Big), \tag{25}$$

so failure probability at most 0.05 requires $N \geq 4\sigma_h^2 z_{0.95}^2 / \|h(\theta_1) - h(\theta_0)\|_2^2 = \Omega((\Delta\eta)^{-6})$.

(iii) Conversely, on the known-midpoint family a linear estimator that projects the averaged impulse response on the normalized third-moment direction and inverts (24) has error at most $|\hat{M}_3 - M_3|/(2p_-p_+(\Delta\eta)^3)$, giving $N = O(\tau^{-2}(\Delta\eta)^{-6}\log(1/\delta_{\text{fail}}))$. On the separated class with weights and rates bounded

Table 9: Two-point sample complexity for two-cohort weight recovery

$\Delta\eta$	ΔM_3	$\|\Delta h\|_2$	N^*_{LeCam} (95%)	N^*_{CRB}
0.30	2.67×10^{-4}	1.98×10^{-3}	7.1	47.8
0.20	7.92×10^{-5}	5.60×10^{-4}	88.2	597.0
0.14	2.72×10^{-5}	1.89×10^{-4}	775.3	5,267.9
0.10	9.90×10^{-6}	6.84×10^{-5}	5,927.6	40,374.2
0.08	5.07×10^{-6}	3.49×10^{-5}	22,740.5	155,048.9
0.06	2.14×10^{-6}	1.47×10^{-5}	128,333.0	875,732.3
0.05	1.24×10^{-6}	8.50×10^{-6}	383,859.1	2,620,321.2
0.04	6.34×10^{-7}	4.35×10^{-6}	1,466,361.5	10,012,662.4

Worst-case failure at $\Delta\eta = 0.10$*,* $N = 50$*, 1,000 trials per hypothesis*: bound 44.1%; ML 94.5%, ridge 47.5%, positive-part shrinkage 100%, heavy prior 73.9%.

$\tau = 0.01$, $\sigma_h = 1.6 \times 10^{-3}$, $a = 0.5$, $\kappa = 0.8$, 121 taps. Recomputed columns from `recompute_theory.py`. Estimators re-implemented in `recompute_p1_estimators.py`: ML is least squares from four starts; ridge and heavy prior are posterior modes with prior standard deviations $(0.05, 0.10)$ and $(0.005, 0.02)$ for (weights, rates) around the symmetric centre of the two hypotheses; positive-part shrinkage pulls the ML weights toward their mean. Failure means some weight is off by at least τ.

away from the boundary, Toeplitz deconvolution followed by a Prony–Hankel root step has Hankel determinant $g_1 g_3 - g_2^2 = w_1 w_2 \eta_1 \eta_2 r_1 r_2 (\Delta\eta)^2 > 0$ *for the deconvolved kernel* $g_m = \sum_i w_i \eta_i r_i^m$*, and perturbation analysis gives the same order.*

We recomputed the two-point quantities exactly (Table 9): the moment gap equals $4\tau p_- p_+ (\Delta\eta)^3$ to machine precision, and the Le Cam requirement at $\Delta\eta = 0.10$ is 5,927.6 resets. The worst-case failure rates of four re-implemented estimators all lie above the lower bound (Table 9).

6.4 Many unknown cohorts

Theorem 6.3 (K-cohort two-point lower bound)**.** *Fix* $K \geq 2$*, a center* $r_c \in (0, 1)$ *and total merged weight* $W > 0$*. There exist two* K*-cohort economies with rates in* $r_c + \Delta\eta[-1, 1]$*, positive weights, and merged measures whose moments of order* $0, \ldots, 2K-2$ *coincide while the moment of order* $2K-1$ *differs by a constant times* $(\Delta\eta)^{2K-1}$*. Their impulse responses differ by* $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$*, so recovering the weights from* N *independent noisy reset experiments with worst-case error probability*

Table 10: Gauss and Gauss–Radau rules on $[-1, 1]$ used in Theorem 6.3

K	Gauss nodes / Gauss–Radau nodes	Max gap, orders $< 2K-1$	Gap at order $2K-1$
2	± 0.5774 / -1, 0.3333	6×10^{-17}	0.2222
3	0, ± 0.7746 / -1, -0.2899, 0.6899	2×10^{-16}	0.0533
4	± 0.3400, ± 0.8611 / -1, -0.5753, 0.1811, 0.8228	4×10^{-16}	0.0131

All Gauss–Radau weights are positive.

Table 11: Numerical check of Theorem 6.3

	$K=2$	$K=3$	$K=4$
Theoretical exponent of N^*	-6	-10	-14
Fitted log–log slope (small $\Delta\eta$)	-5.96	-9.96	-14.01

Le Cam bound $N^* \propto \sigma^2/\|h^A - h^B\|_2^2$ with $\sigma = 1$, $a = 0.5$, $\kappa = 1$, total merged weight 0.10, rates centered at 0.70, responses over 400 lags.

below 1/4 *requires*

$$N \;\geq\; c\,\sigma^2(\Delta\eta)^{-(4K-2)}.$$

The construction uses the K-point Gauss and Gauss–Radau rules of the same weight function: both are positive, have K atoms and integrate polynomials of degree $2K-2$ exactly. For $K = 2$ it reproduces the order of Theorem 6.2. Table 10 lists the two rules: their low moments agree to machine precision and their moment of order $2K-1$ differs by 0.222, 0.0533 and 0.0131 for $K = 2, 3, 4$. The shrinking gap explains why constants deteriorate with K on top of the exponent.

Table 11 and Figure 6 confirm the order numerically.

Theorem 6.4 (Matching upper bound for every K)**.** *Fix $K \geq 2$, a compact set $\mathcal{S} \subset [-1, 1]^K$ of rescaled rates with $\min_{i \neq j} |s_i - s_j| \geq \gamma > 0$, and weights in $[\omega_{\min}, \omega_{\max}]$ with $\omega_{\min} > 0$. Consider the class of K-cohort economies with rates $r_c + \Delta\eta\, s$, $s \in \mathcal{S}$, and N independent reset experiments whose responses carry Gaussian noise with variance σ^2. There are constants $c_0, C > 0$ depending only on $(K, \mathcal{S}, \gamma, \omega_{\min}, \omega_{\max}, a, \kappa, r_c)$ such that, whenever $\epsilon_N \equiv \sigma N^{-1/2}(\Delta\eta)^{-(2K-1)} \leq c_0/\sqrt{\log(2/\beta)}$, Prony's method applied to*

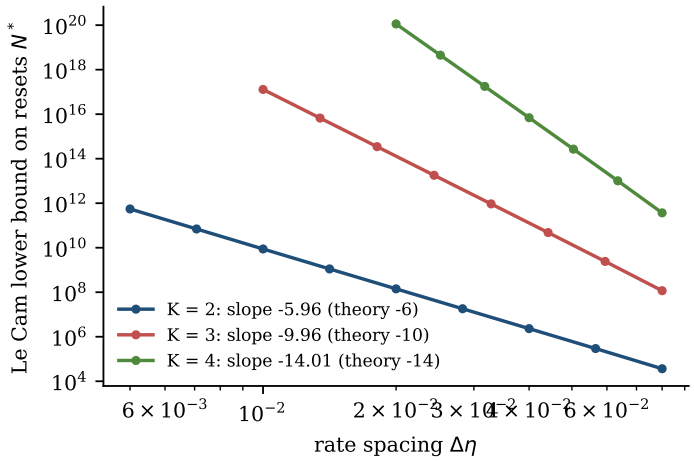


Figure 6: Two-point lower bound on the number of reset experiments needed to separate K-cohort economies, as a function of the rate spacing.

the first $2K$ estimated kernel values recovers the weights with

$$\max_i |\hat{w}_i - w_i| \;\le\; C\sqrt{\log(2K/\beta)}\,\epsilon_N$$

with probability at least $1-\beta$. Hence $N = C'\sigma^2(\Delta\eta)^{-(4K-2)}$ experiments suffice for any fixed accuracy and confidence, and together with Theorem 6.3 the minimax sample requirement is $N^* \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$.

Proof. The kernel values $k_m = \sum_i w_i r_i^m$ are the power moments of the merged measure. The closed-loop recursion $h_t = ah_{t-1} + \kappa\sum_m k_m h_{t-1-m}$ (plus κ at $t = 0$) can be solved for $k_0, \dots, k_{2K-1}$ from $h_0, \dots, h_{2K}$ with $h_0 = \kappa \neq 0$; the map is smooth with Lipschitz constant independent of $\Delta\eta$, so the estimated kernel values have error $O_P(\sigma N^{-1/2})$. Centring at r_c is a fixed triangular linear map, $\mu_m = \sum_{j\le m}\binom{m}{j}(-r_c)^{m-j}k_j$, and $\mu_m = (\Delta\eta)^m t_m$ with $t_m = \sum_i w_i s_i^m$. The map $F : (s,\omega) \mapsto (t_0, \dots, t_{2K-1})$ does not depend on $\Delta\eta$; its Jacobian is a confluent Vandermonde matrix with determinant $\prod_i \omega_i \prod_{i<j}(s_i - s_j)^4$, bounded away from zero on the compact class. By the quantitative inverse function theorem there are $\rho_0 > 0$

Table 12: Sample requirement of Prony's method against the lower-bound order

	$K=2$	$K=3$	$K=4$	$K=5$
Lower-bound order $-(4K-2)$	-6	-10	-14	-18
Fitted slope of N^* in $\Delta\eta$ (Prony)	-5.94	-10.00	-13.93	-18.34

Rates $0.7 + \Delta\eta\, s$ with s the K-point Gauss nodes, total weight 0.1, $\Delta\eta \in [0.03, 0.2]$; N^* is the number of experiments for a weight RMSE of 10% of the smallest weight, from 400 simulated estimates per point.

and $L < \infty$, uniform over the class, such that F^{-1} exists and is L-Lipschitz on the ball of radius ρ_0 around every point of $F(\text{class})$. An error e in μ_m is an error $e(\Delta\eta)^{-m}$ in t_m, largest at $m = 2K-1$. The kernel errors are Gaussian with standard deviation of order $\sigma N^{-1/2}$, so a union bound over the $2K$ coordinates gives $\|\hat{t} - t\|_\infty \le C_1\sqrt{\log(2K/\beta)}\,\epsilon_N$ with probability at least $1-\beta$. When this is below ρ_0, $\hat{t}$ lies in the ball where F^{-1} is L-Lipschitz. Prony's method computes the algebraic inverse of the moment map, which does not depend on coordinates, so its output is $F^{-1}(\hat{t})$ and the stated bound follows with $C = LC_1$ and $c_0 = \rho_0/C_1$. The Gauss and Gauss–Radau pair of Theorem 6.3 lies in the class, so the orders match. □

Table 12 confirms the order numerically. For numerical stability the estimator works in centred and rescaled coordinates (the estimator is the same; only its floating-point evaluation changes). The argument parallels the stability analysis of super-resolution for clustered complex nodes (Batenkov et al., 2021), here for real nodes, and it reproduces the ESPRIT slope of -6.15 reported for $K = 2$ in the earlier version.

6.5 General memory: the minimax rate is logarithmic

Exponential memory is a modelling choice. Suppose instead that the feedback kernel is completely monotone, $k_m = \int_{[0,\rho]} r^m\, d\mu(r)$ with $\rho < 1$, where μ is any nonnegative

measure of total mass W: a finite or infinite mixture of exponential memories with merged weight W. Write $\mathcal{M}(\rho, W)$ for this class, assume the closed loop is uniformly stable, $s \equiv a + \kappa W/(1-\rho) < 1$, and measure accuracy by

$$d(\mu, \mu') = \sup\Big\{ \int f \, d(\mu - \mu') : \ f \text{ 1-Lipschitz on } [0, \rho], \ f(0) = 0 \Big\},$$

which equals the Wasserstein-1 distance between measures of equal mass and also compares estimates whose mass is not exactly W. Let $E_L(|x|)$ denote the error of the best uniform approximation of $|x|$ on $[-1, 1]$ by polynomials of degree L, and let $c_* = \inf_{L \geq 1} L\, E_L(|x|)$. Bernstein (1914) showed that $L\, E_L(|x|)$ converges to a positive constant (0.2802; Varga and Carpenter, 1985), so $c_* > 0$; numerically $c_* = 1/4$, attained at $L = 2$ (Table 13).

Lemma 6.5 (Moment-matched cohort pairs)**.** *For every $n \geq 2$ there are two probability measures on $[0, \rho]$, each with exactly n atoms, whose moments of orders $0, \ldots, 2n-2$ coincide and whose Wasserstein-1 distance is at least $\rho\, E_{2n-2}(|x|) \geq c_* \rho/(2n-2)$.*

The atoms are the points where the best approximation of $|r - \rho/2|$ by polynomials of degree $2n-2$ attains its maximal error with alternating signs, and the weights are those of the divided difference on these points, which annihilates every polynomial of degree $2n-2$ (Appendix A). Each measure is an n-cohort economy.

Theorem 6.6 (Minimax rate for general memory)**.** *Let N independent reset experiments observe the impulse response with Gaussian noise of variance σ^2 per tap, and let $B(\rho) = (1+\sqrt{2})(1+2/\rho)$ and $C_h = \kappa^2/(1-s)^2$.*

(i) Lower bound. *Let $\Lambda_N = \log(N/\sigma^2) + \log\big(2C_h^2 W^2/((1-\rho)\rho)^2\big)$. Whenever*

$\Lambda_N \geq 4\log(1/\rho)$, *every estimator* $\hat{\mu}$ *satisfies*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu) \geq \frac{c_*\,\rho W\log(1/\rho)}{\Lambda_N}\Big) \;\geq\; \frac{1}{4e},$$

and the bound already holds over n*-cohort economies, for a finite* n *that grows with* N*.*

(ii) Upper bound. *With* a *and* κ *known, invert the averaged response into kernel values* $\hat{k}_0,\dots,\hat{k}_L$*, with* $L+1 = \lfloor \log(N/\sigma^2)/(4\log B(\rho))\rfloor$*, and let* $\hat{\mu}$ *be any measure on* $[0,\rho]$ *minimizing* $\max_{j\leq L}|\int r^j d\hat{\mu} - \hat{k}_j|$ *(a linear program). For every* $\beta\in(0,1)$ *there is* N_0 *depending only on* (ρ,W,a,κ,β) *such that for* $N/\sigma^2 \geq N_0$*,*

$$\sup_{\mu\in\mathcal{M}(\rho,W)} P_\mu\Big(d(\hat{\mu},\mu) > \frac{3\pi\rho W\log B(\rho)}{\log(N/\sigma^2)}\Big) \;\leq\; \beta.$$

Hence the minimax accuracy is of exact order $1/\log(N/\sigma^2)$*, and reaching accuracy* w *requires and suffices* $N/\sigma^2 = \exp(\Theta(1/w))$*.*

Proof sketch. (i) Put the measures of Lemma 6.5, scaled to mass W, into the model. Their difference annihilates polynomials of degree $2n-2$, so the generating function of the kernel difference is $\int(1-rz)^{-1}d(\mu_A-\mu_B) = z^{2n-1}\int r^{2n-1}(1-rz)^{-1}d(\mu_A-\mu_B)$; on $|z|=1$ it is at most $2W\rho^{2n-1}/(1-\rho)$, and Parseval bounds $\|k^A-k^B\|_2$ by the same quantity. The response difference is H_AH_B times the shifted kernel difference, so $\|h^A-h^B\|_2 \leq C_h\|k^A-k^B\|_2$. Choosing the smallest n for which the Kullback–Leibler divergence of the N-sample experiments is at most one, the Bretagnolle–Huber form of Le Cam's lemma gives the bound with half the Wasserstein separation of the lemma. (ii) For a 1-Lipschitz f with $f(0)=0$, Jackson's theorem gives a polynomial q of degree L with $\sup|f-q| \leq \pi\rho/(4(L+1))$, and expanding q in Chebyshev polynomials of $2r/\rho-1$ bounds the sum of its monomial coefficients by

$\rho B(\rho)^{L+1}$. Hence $d(\hat{\mu}, \mu) \le \pi\rho(W + \delta)/(2(L + 1)) + 2\delta\rho B(\rho)^{L+1}$, where δ is the largest kernel error. The kernel is recovered from the response by a series reciprocal, $K(z) = (1 - az - \kappa/H(z))/(\kappa z)$, whose error is at most $2((1 + s)/\kappa)^2(L + 2)$ times the largest response error, so $\delta = O_P(\sigma N^{-1/2} L\sqrt{\log L})$. The choice of L makes $\delta B(\rho)^{L+1} = O_P((\sigma^2/N)^{1/4} L\sqrt{\log L})$, negligible against the first term. Full proofs are in Appendix A. □

Theorem 6.6 replaces a polynomial barrier with a logarithmic one. With exponential memory and K known cohorts, halving the error multiplies the required number of experiments by a constant; with general memory, halving the error squares it. The lower bound holds already for finite mixtures, so knowing that the memory is a finite mixture of exponentials does not help unless the number of cohorts and their spacing are also known. The constants of the two bounds differ, by a factor that grows as $\rho \to 1$, so the dependence on how long memory lasts is not pinned down. The source of the gap is visible in the proof: as $\rho \to 1$ the moment-matched pairs separate their kernels more slowly (ρ^{2n-1} decays slowly), which weakens the lower bound through $\log(1/\rho)$, while the upper bound pays $\log B(\rho)$, which stays bounded away from zero, for the conditioning of polynomial approximation on $[0, \rho]$. Logarithmic rates of this kind are familiar from deconvolution with supersmooth errors (Fan, 1991) and from moment methods for mixing distributions (Kong and Valiant, 2017; Wu and Yang, 2020); here they arise from the closed loop of a market.

Table 13 checks the construction and the estimator. The pairs of Lemma 6.5 have exactly n atoms, agree in all moments below order $2n - 1$ to 10^{-61}, and remain at Wasserstein distance about $0.4/n$, while the distance between their kernels shrinks by a factor of about 0.2 per atom at $\rho = 0.9$. The linear-programming estimator applied to a continuous memory distribution has error times $\log(N/\sigma^2)$ between 0.18 and 0.28

Table 13: General memory: moment-matched cohort pairs and the moment estimator

A. Best approximation of $\lvert x\rvert$						
L	2	4	6	10	20	30
$L\,E_L(\lvert x\rvert)$	0.2500	0.2705	0.2756	0.2785	0.2797	0.2800
B. Pairs of Lemma 6.5						
Atoms per measure n	2	4	6	8	10	12
$W_1(\mu_A,\mu_B)$, $\rho=0.9$	0.2250	0.1037	0.0669	0.0494	0.0391	0.0324
$n\,W_1$, $\rho=0.9$	0.450	0.415	0.402	0.395	0.391	0.388
$\lVert k^A-k^B\rVert_2$, $\rho=0.9$	4.41×10^{-2}	1.03×10^{-3}	4.08×10^{-5}	1.96×10^{-6}	1.05×10^{-7}	5.95×10^{-9}
$\lVert k^A-k^B\rVert_2$, $\rho=0.99$	7.41×10^{-2}	5.84×10^{-3}	1.07×10^{-3}	2.76×10^{-4}	8.33×10^{-5}	2.76×10^{-5}
C. Linear-programming estimator, continuous memory distribution, $\rho=0.9$						
$\log_{10} N$	4	8	12	16	20	24
Moments used L	2	3	5	6	7	8
Error $d(\hat\mu,\mu)$	0.00878	0.00759	0.00432	0.00445	0.00382	0.00399
Error $\times\log(N/\sigma^2)$	0.202	0.245	0.179	0.226	0.229	0.276

A: Remez algorithm; the limit is Bernstein's constant 0.2802. B: moments of orders $0,\dots,2n-2$ agree to 10^{-61} (60-digit arithmetic); every W_1 exceeds $\rho E_{2n-2}(\lvert x\rvert)$. C: true merged measure Beta(2,3) on $[0,0.9]$ with mass 0.1; kernel values observed with noise $\sigma N^{-1/2}$, $\sigma=10^{-3}$; L from the rule in Theorem 6.6(ii); measure fitted on a 401-point grid; 20 replications. Kind: recomputed (`general_minimax.py`).

for N from 10^4 to 10^{24}: a 10^{20}-fold increase in data halves the error. The exponential-memory model of the previous subsections, where the cost is polynomial in $1/\Delta\eta$, is therefore the optimistic case: relaxing the parametric form makes identification from prices much harder, not easier.

6.6 Passive data

Theorem 6.7 (Passive identification: minimum phase)**.** *In any stable economy with $\lvert r_i\rvert<1$ and nonzero merged weights, the zeros of H are exactly the memory rates r_i, and H is minimum phase. With a single price innovation, the price spectrum identifies H up to scale, hence the memory rates; with κ and a known, as in Theorem 3.1, it identifies the merged measure.*

The result holds for signed weights (mean-reverting cohorts): in 1,733 stable random economies with weights of both signs, the computed zeros equalled the memory rates and lay inside the unit disk. With a second independent shock in the cohorts'

memories, the Jacobian of the price spectrum with respect to six model parameters has full rank, with singular values from 34.51 down to 0.0698. Passive noise helps identification rather than hindering it.

Theorem 6.8 (Passive cost)**.** *Suppose the price is observed passively as the stationary output $x_t = H_\theta(L)\xi_t$ of a single innovation with variance $\sigma^2 > 0$, for θ in a compact, uniformly Schur-stable family with rates in $[\eta_{\min}, \eta_{\max}] \subset (0, 1)$, and let $D(\theta_1 \| \theta_0)$ be the Kullback–Leibler divergence rate per observation.*

(i) For a single cohort at $\bar{\eta}$ against the symmetric pair at $\bar{\eta} \pm \delta$ with the same total weight, $D = \Theta(\delta^4)$.

(ii) For the moment-matched two-cohort economies of Theorem 6.2, $D = \Theta(\tau^2(\Delta\eta)^6)$.

Hence any test based on T passive observations with worst-case error at most $\alpha < 1/4$ needs $T = \Omega(\delta^{-4})$ and $T = \Omega(\tau^{-2}(\Delta\eta)^{-6})$ respectively, and the likelihood-ratio test attains these orders as $T \to \infty$.

The proof (Appendix A) combines the Whittle form of the divergence rate with the resolvent identity $H_1 - H_0 = H_1 H_0 (G_1 - G_0)$: the two spectra differ by the leading unmatched moment, of order δ^2 for the pair and $\tau(\Delta\eta)^3$ for the moment-matched economies, and the divergence rate is the square of that difference. Numerically, the fitted slopes of the required sample length are -4.21 and -5.93, against the theoretical -4 and -6.

Passive surveillance is possible in principle but costs the same polynomial order as active probing. Proposition 4.6 is the exact finite-sample version for the known pair.

Table 14: Disclosure contracts at $\delta = 0.05$ (pulse-design capital, par units)

Contract	Channel	Pulses	Benchmark	Saving vs. $T = 20$
I_0: prices, $T = 3$	public prices	271	4,305.25	−2,123.5%
I_0^*: prices, $T = 20$	full price path	12	193.62	0.0%
I_1: prices and signed flow	public flow	12	193.62	0.0%
I_2: noisy equity ($\sigma_E = 0.5$)	statement	12	193.62	0.0%
I_3: exact equity, 5-unit lot	audited equity	1	12.70	93.4%
I_3: exact equity, 0.01 rounding	rounded report	1	15.52	92.0%
I_4: timestamps only	no likelihood	12	193.62	0.0%
I_5: audit registry	internal states	0	0.00	100.0%

Closed-form values. Noisy equity adds information linearly in the number of resets, $D_{\mathrm{KL}} = N\Delta_E^2/(2\sigma_E^2)$, and saves only 1.5% against the $T = 3$ baseline. The equity difference is $\Delta E = \kappa^2 q \delta^2 V^2$; with reports rounded to 0.01, a guaranteed change needs $V \geq 5.59$.

7 What disclosure buys

Because active probing is costly and passive data are slow, the natural policy question is what a supervisor gains from disclosure. We compare six observation contracts at $\delta = 0.05$.

- I_0: public prices only.
- I_1: prices and aggregated executed flow.
- I_2: prices and one independent noisy report of dealer marked equity per reset ($\sigma_E = 0.5$).
- I_3: prices and exact audited equity (Proposition 4.4), also with reports rounded to 0.01.
- I_4: algorithm update timestamps without the update values.
- I_5: a supervisory audit registry with access to internal algorithm states.

Table 14 shows that disclosure is not one thing. Timestamps without update values carry no information about δ in this model. Noisy periodic statements barely help. Exact audited equity collapses the requirement to a single trade, limited only by

lot sizes and report precision, and an audit registry removes the trading requirement altogether.

The cleanest form of useful disclosure is flow tagged by participant class.

Proposition 7.1 (Tagged flow removes the barrier)**.** *If the observer receives each cohort's flow $u_{i,t} = q_i y_{i,t}$ with measurement noise, the merged weights are estimated at rate $T^{-1/2}$ with a constant that does not depend on the rate spacing.*

Each tagged flow follows $u_{i,t} = r_i u_{i,t-1} + w_i x_t$, a regression whose information about w_i does not involve the other cohorts' rates. In simulation ($T = 5{,}000$, 60 replications) the root-mean-square error of the recovered weights is flat in $\Delta\eta$ (fitted slope 0.011, error between 5.8×10^{-4} and 6.6×10^{-4}). This is the formal content of a disclosure regime such as a consolidated audit trail: tagging flow by participant class turns a polynomial barrier into an ordinary estimation problem.

7.1 How much disclosure is worth

Proposition 7.1 says that tagging removes the barrier; the next result says by how much.

Theorem 7.2 (Value of class-tagged disclosure)**.** *In the class of Theorem 6.4, let N_p be the number of experiments needed with prices alone and N_t the number needed with each class's flow observed with noise σ_f. To recover the weights to fixed accuracy, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-2)}$; to resolve the rates to accuracy proportional to their spacing, $N_p/N_t \asymp (\sigma/\sigma_f)^2(\Delta\eta)^{-(4K-4)}$.*

Proof. $N_p \asymp \sigma^2(\Delta\eta)^{-(4K-2)}$ by Theorems 6.3 and 6.4. With tagged flow each class is a single-exponential problem whose Fisher information for (w_i, r_i) does not depend on the other classes, so $N_t \asymp \sigma_f^2$ for the weights and $N_t \asymp \sigma_f^2(\Delta\eta)^{-2}$ for rates resolved to a fraction of $\Delta\eta$. □

With the same noise and lags for both observers, the Cramér–Rao requirements give fitted slopes of -6.08 for prices and -4.34 for the ratio at $K = 2$ (theory -6 and -4). Calibrated to the class memories estimated below, prices alone would need about 333 times as many days as the published class ratios to separate the top-trader and all-account memories in BTCUSDT (rates 0.9889 and 0.9986 per five minutes; about 6,000 days against 18), and about 19 times as many in ETHUSDT (rates 0.9793 and 0.9972).

7.2 Disclosure in real data

Binance publishes, every five minutes, long/short ratios for its top-trader group, by position and by account, and for all accounts. These are class-level positions, the closest public analogue to the tagged data of Proposition 7.1. We treat a class's reported net position as the observable counterpart of its cohort state: it obeys the same one-lag recursion class by class, so the block-diagonal information structure of Proposition 7.1 carries over. The mapping is an assumption, because the reports are positions rather than flows and returns stand in for the unobserved deviation x_t. For each class c we estimate $n_{c,t} = r_c n_{c,t-1} + w_c \,\mathrm{ret}_t$, with $n = (R-1)/(R+1)$ from the ratio R. The price-only alternative regresses signed five-minute taker flow on twelve lagged five-minute returns and fits two exponentials to the kernel; it estimates the memory rates without class information. Intervals come from a day bootstrap with 200 draws.

Table 15 gives three results. First, tagged class data pin down memory rates that prices and flow leave essentially unrestricted: the price-only intervals are 28 to 124 times wider in every one of 30 symbols on fresh days, and cover most of the admissible range. Part of this ratio is mechanical, because a series of positions identifies its own

Table 15: Memory rates with and without class-tagged data

Symbol	Days	Top position $\hat{r}$ [95% CI]	All accounts $\hat{r}$ [95% CI]	Width ratio	Classes differ
BTCUSDT	48	0.9889 [0.9842, 0.9935]	0.9986 [0.9955, 1.0005]	121.7	yes
ETHUSDT	48	0.9793 [0.9616, 0.9907]	0.9972 [0.9934, 0.9997]	55.6	yes
CELRUSDT	48	0.9884 [0.9830, 0.9922]	0.9882 [0.9804, 0.9914]	81.6	no
CVXUSDT	33	0.9822 [0.9732, 0.9908]	0.9829 [0.9694, 0.9900]	45.9	no

Pre-registered test on fresh days, 30 USD-M perpetuals, 2025–2026	
(i) width ratio > 10	30 of 30 (range 27.6–124.0)
(ii) price-only rate interval wider than 0.25	30 of 30 (range 0.65–0.99)
(iii) merged response $\sum_m k_m$ identified	5 of 30
Classes differ	11 of 30

Top panel: training days 2023–2024; width ratio is the mean price-only rate-interval width over the mean tagged width. The price-only intervals for both rates cover most of $[0, 0.99]$. Bottom panel: specification registered before the fresh days were downloaded (SHA-256 `65acda361861...`); a symbol passes if all three of the first rows hold, and the test required 80% of symbols to pass.

persistence directly while the price-only route must fit two exponentials to a weakly identified kernel. Restricting attention to the five symbols in which the price-only fit does identify the merged response (ACH, API3, OGN, ZEN and DASH) does not change the conclusion: the price-only rate intervals are still 0.64 to 0.92 wide and the ratios run from 27.6 to 62.5 (median 47.5). We therefore read the ratios as a measure of how little prices reveal about class memory, not as a horse race between two estimators of equal standing. Second, the classes differ where theory says the information should be: in BTCUSDT and ETHUSDT the top-trader group's positioning has a half-life of about three to five hours against about one to two days for all accounts, while in thin symbols the classes are often indistinguishable. Third, the pre-registered test fails its 80% criterion, on the part of the prediction that concerns what prices *do* reveal: the merged response is significantly different from zero in only 5 of 30 symbols. Theorem 6.7 says the merged measure is identified; on 42 days of thin-market data it is not estimated precisely. Prices reveal even less than the identification theory guarantees, which strengthens rather than weakens the case for class-tagged disclosure.

Class positions do not help forecast short-horizon risk: adding the level and one-hour change of the top-trader and all-account positions to the controls of the companion paper lowers out-of-sample tail classification and leaves R^2 unchanged (-0.19 points, CI $[-0.99, 0.83]$). Their value is in identifying who trades on what memory, not in timing the next hour.

7.3 A second disclosure regime: the CFTC's trader-class reports

Binance publishes its class ratios voluntarily, for one venue. The U.S. Commodity Futures Trading Commission requires large traders in regulated futures markets to report their positions and publishes them weekly by class: leveraged funds, asset managers and dealers in financial futures, and managed money, producer/merchants and swap dealers in commodities. This is class-tagged disclosure mandated by law. We registered a test (SHA-256 `6f1d6f35ae88...`) before downloading any data, in eight markets that span asset classes: CME Bitcoin and Ether, the E-mini S&P 500, 10-year Treasury notes, the euro, the yen, WTI crude oil and Henry Hub natural gas, using every weekly report through October 2026 (282 to 1,039 weeks per market). For each class, $n_{c,t} = (\text{long} - \text{short})/\text{open interest}$ follows $n_{c,t} = \alpha_c + r_c n_{c,t-1} + w_c \, \text{ret}_t$, the weekly version of the cohort recursion. The price-only alternative fits a two-exponential kernel to twelve weekly return lags. Intervals come from a 13-week moving-block bootstrap. The registered hypotheses were: (H1, primary) the fast class's memory rate is pinned to an interval narrower than 0.25 while both price-only intervals are wider, in at least 75% of markets; (H2) the fast class trend-follows, $w_c > 0$, in at least 75%; (H3) the fast class has the shorter memory in at least half.

The primary hypothesis holds in all eight markets (Table 16). The class reports

Table 16: Trader-class memory from the CFTC Commitments of Traders, eight regulated futures markets

Market	Weeks	Fast $\hat{r}$ [95% CI]	Slow $\hat{r}$	$\hat{r}_{\text{fast}} - \hat{r}_{\text{slow}}$ [95% CI]	$\hat{w}_{\text{fast}}$	Ratio
Bitcoin (CME)	434	0.951 [0.928, 0.969]	0.986	−0.035 [−0.062, −0.010]	−	23.7
Ether (CME)	282	0.890 [0.826, 0.932]	0.960	−0.070 [−0.143, −0.011]	−	9.1
E-mini S&P 500	1,039	0.914 [0.884, 0.937]	0.954	−0.040 [−0.071, −0.012]	−	18.3
10-year T-note	1,039	0.992 [0.985, 0.998]	0.993	−0.000 [−0.006, +0.007]	(0)	71.4
Euro	1,035	0.961 [0.947, 0.972]	0.996	−0.034 [−0.051, −0.022]	+	37.3
Yen	1,035	0.955 [0.939, 0.968]	0.984	−0.029 [−0.047, −0.013]	+	34.0
WTI crude oil	1,039	0.961 [0.946, 0.974]	0.999	−0.037 [−0.053, −0.023]	+	35.2
Natural gas	1,039	0.981 [0.969, 0.990]	0.980	+0.001 [−0.014, +0.017]	+	44.0

Pre-registered hypotheses: H1 (primary) 8 of 8, pass; H2 4 of 8, fail; H3 6 of 8, pass.

Fast class: leveraged funds (financial futures) or managed money (commodities); slow class: asset managers or producer/merchants. $\hat{w}_{\text{fast}}$: sign of the response to returns, + or − when the 95% interval excludes zero, (0) otherwise. Ratio: mean width of the two price-only rate intervals over the width of the fast-class interval; the price-only intervals span 0.95–0.96 of the admissible range $[0, 0.99]$ in every market. Weekly returns from Binance spot (Bitcoin, Ether) and Yahoo Finance (others; the Treasury return is minus the change in the 10-year yield). Specification SHA-256 `6f1d6f35ae88...`; test opened once. Kind: real data (Part 11).

pin each fast class's memory rate to an interval between 0.013 and 0.106 wide, while weekly prices leave both memory rates anywhere in the admissible range; the width ratios run from 9 to 71. The ratio partly reflects how little weekly returns reveal, which is the point of the theory rather than an artefact of it: with prices alone the memories are not located at all. The classes differ as the cohort model allows. Leveraged funds and managed money have the shorter memory in six of eight markets, with half-lives of 6 to 18 weeks against months to years for asset managers and producers; in Treasuries and natural gas the two memories are indistinguishable.

The trend-following hypothesis fails. The fast class buys after rises in the euro, the yen, crude oil and natural gas, but sells after rises in Bitcoin, Ether and the S&P 500 futures. The first half matches the finding of Moskowitz et al. (2012) that speculators' positions tend to go with past returns. We did not predict this. One candidate explanation is that leveraged-fund positions in financial futures are dominated by the short futures leg of cash–futures basis trades, which a single memory per class

cannot represent. A post-hoc check is consistent with it but not decisive (Part 13, R6): weekly position changes of leveraged funds and asset managers are strongly negatively correlated in Bitcoin, Ether, the S&P 500 and Treasuries (-0.37 to -0.67) but not in the euro or yen (-0.04 and -0.05), and holding the asset managers' change fixed removes the negative response in Bitcoin and halves it in Ether and the S&P 500. Because class positions sum to zero across all reporting categories, some negative correlation is mechanical, so this is a signature, not a test. The failure sharpens rather than weakens the case for class tagging: class labels are not strategies, and a supervisor who wants to know how a class responds to prices needs its positions, which the price process does not reveal.

Together with Table 15, the evidence now covers two disclosure regimes, one voluntary and one mandated by law, in crypto and in regulated equity-index, rate, currency and commodity futures. In both, class-tagged positions locate memories that prices leave unrestricted.

8 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem 8.1 (Spectral blindness)**.** *For any eigenvalues $0 < \lambda_2 < \lambda_1 < 1$, one-step price variance $s_1 > 0$ and stationary price variance V large enough, there is a family of two-state markets $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ with shock covariance $\mathrm{diag}(s_1, s_2(\theta))$ such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain $\sup_k \|e_1^\top J_\theta^k\|$ grows without bound in θ.*

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative

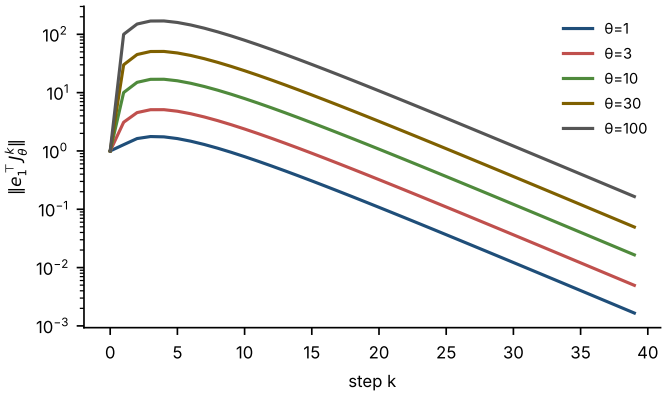


Figure 7: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure 7).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.4). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix C.4). Second, in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of Lemma 2.4; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator (`recompute_oct20.py`). Non-normality is necessary for transient amplification of the price, not sufficient; where amplification comes from is the subject of the companion paper.

9 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 7.2 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

9.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{26}$$

Proposition 9.1 (Mechanical persistence under filtered random walks)**.** *If p_b is a martingale, $\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}}$: the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.*

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{27}$$

Table 17: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{28}$$

Proposition 9.1 implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

9.2 Data

9.3 Clock alignment dominates reduced-form feedback

We estimate (27) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table 18), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical

Table 18: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

Table 19: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

value (0.9351) with $\hat{Q}$ indistinguishable from zero.

9.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table 20 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small,

Table 20: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table 18.

Table 21: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table 18; the placebo shows only that the filter does not produce it.

9.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta \Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table 21).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (Part 13, `R7`, post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table 19 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table 19, column $\hat{Q}_{30}$), consistent with persistent order splitting.

9.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

Table 22: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

9.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table 22).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is -0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

10 Implications for supervision and market design

What a supervisor can learn from public data. Prices and anonymous flow reveal the merged measure M, the closed-loop impulse response and the reduced-form operator, and nothing finer. A supervisor who wants to know *which* algorithms drive a feedback loop cannot get there from public data at reasonable cost when their memories are close. This is not a failure of econometric technique: Theorem 6.2 holds for every estimator, biased or not.

What probing costs. At realistic observation noise, an exchange or a large desk could separate two well-separated memory rates for about 10^5 USDT of capital. Probing is therefore feasible for coarse questions and infeasible for fine ones. Because capital grows as δ^{-4} and experiments as $(\Delta\eta)^{-(4K-2)}$, halving the separation multiplies the cost by 16 for a known pair and by 2^{4K-2} for K unknown cohorts.

What disclosure buys. Disclosure regimes differ sharply in value (Table 14). Timestamps without update values are worthless for this purpose. Noisy periodic statements are nearly worthless. Audited equity and, above all, flow tagged by participant class remove the barrier entirely (Proposition 7.1). This gives a quantitative argument for audit-trail regimes that record the participant class of each order, and a quantitative argument against relying on periodic statements for the same purpose. Calibrated to Binance's published class ratios, separating the memories of the top-trader group and of all accounts in BTCUSDT would take prices alone about 333 times as long (Theorem 7.2). The class reports that the CFTC already requires have the same property in regulated markets (Table 16), so the argument concerns the granularity and frequency of existing disclosure, not a new kind of reporting.

When audit trails already exist. The US Consolidated Audit Trail (SEC Rule 613) gives the SEC and the self-regulatory organizations a record of every order in NMS stocks and listed options from origination to execution or cancellation, linked to an anonymized customer identifier and to the broker-dealer that handled it; since 2025 it no longer collects the names, addresses and years of birth of natural persons (SEC, 2012, 2025). Our results bear on such regimes in three ways. First, they price what the record is worth for the question of who drives a feedback loop: participant-level records remove a barrier that otherwise costs capital of order δ^{-4} and samples of order $(\Delta\eta)^{-(4K-4)}$ (Proposition 7.1, Theorem 7.2). For this purpose a stable account identifier is what matters and names are not needed, which is a quantitative reason to keep the account linkage while removing personal data. Second, an audit trail records accounts, not decision rules. One account can run several algorithms and one algorithm can trade through several accounts, so an observer with account-level records who wants rule-level memories faces our problem within each account's flow, with fewer cohorts but the same exponent; MiFID II's algorithm identifier (ESMA, 2016) is the closer analogue of the tagged flow in Proposition 7.1. Third, the audit trail does not reach the markets we study. Crypto perpetual futures are outside it, futures regulated by the CFTC have separate reporting, and its data are not available to exchanges, other participants or researchers. For these observers the price-only and coarse-disclosure contracts of this paper are the relevant ones.

What to monitor instead. Because transient loss can be arbitrarily large at fixed eigenvalues and volatility (Theorem 8.1), supervision that wants to control risk rather than identify participants should monitor quantities that bound transient loss directly: the summed absolute impulse response and operator confidence sets, both of which are identified from public data. The companion paper develops this.

How to measure feedback. Empirical feedback indices should (i) compare persistence with the mechanical value of the filter, (ii) align flow with the bar in which it is observed, (iii) control for order-splitting persistence or use instruments with a defended exclusion, and (iv) report placebo pipelines. Each of these changes the sign or the order of magnitude of published-style estimates in our data.

11 Conclusion and an open problem

Prices reveal a merged summary of algorithmic feedback. Recovering the algorithms behind it costs executed quantity of order δ^{-2} and trading capital of order δ^{-4} for a known pair, with a certified constant of about 61 par units at the calibration, and a number of experiments of exact order $(\Delta\eta)^{-(4K-2)}$ for K unknown cohorts, passively as well as actively. Without the exponential form, accuracy improves only as $1/\log N$, with matching lower and upper bounds. Class-tagged disclosure removes the barrier, and its value can be stated in samples: a factor of order $(\Delta\eta)^{-(4K-4)}$, about 333 for the trader classes Binance already publishes in BTCUSDT; the CFTC's class reports locate memories that prices leave unrestricted in all eight regulated markets we examined. At realistic noise the capital barrier for a moderate separation is about 10^5 USDT. Measured feedback in crypto markets is real but small, rises with depth, does not move when fees change trading volume severalfold, and is easily confused with filtering and timing artefacts. Finally, eigenvalues and volatility cannot certify safety, because transient loss can be arbitrarily large at fixed spectrum and variance.

This leaves an open question for supervision: *what can be measured and controlled without identifying the algorithms?* The impulse response $g_m = c^\top J^m b$, the operator J and finite-sample confidence sets for it are identified from public data even when cohorts are not. The companion paper uses exactly this interface to build an

identification-free pre-trade control.

Data and code availability. All data are public: Binance archives (https://data.binance.vision), verified against published checksums. The pre-registered specification of the disclosure test, its SHA-256 hash and its test-set unlock counter are in the replication package. Every number in this paper is produced by a notebook cell or script that writes a result file (Appendix D).

Use of AI tools. The authors used Claude (Anthropic), an AI assistant, to write and debug analysis code, run checks of derivations and numerical results, and draft and edit text. Every number in the paper is produced by code the authors executed and archived in the replication package. The authors reviewed all content and take full responsibility for it.

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A Proofs

Proposition 2.1. For $v_{t+1} = v_t + \zeta_{t+1}$ and $r_t = v_t + \nu_t$, the steady-state prior variance P solves $P = P\sigma_\nu^2/(P+\sigma_\nu^2) + \sigma_\zeta^2$, i.e. $P^2 - \sigma_\zeta^2 P - \sigma_\zeta^2\sigma_\nu^2 = 0$, whose positive root is $P = \big(\sigma_\zeta^2 + \sqrt{\sigma_\zeta^4 + 4\sigma_\zeta^2\sigma_\nu^2}\big)/2$. The gain is $\eta = P/(P+\sigma_\nu^2) \in (0,1)$, which simplifies to (6), and the update $\hat{v}_{t+1} = \hat{v}_t + \eta(r_t - \hat{v}_t)$ has the stated form.

Proposition 2.2. $\nabla_\theta \frac{1}{2}H(x_t-\theta)^2 = -H(x_t-\theta)$, so $\theta_{t+1} = (1-\gamma)\theta_t + \alpha H(x_t-\theta_t) = (1-\eta)\theta_t + \eta q x_t$ with $\eta = \gamma + \alpha H$ and $q = \alpha H/\eta$.

Lemma 2.4. Omit zero-weight cohorts. For $a > 0$ put $b_i = \kappa q_i$ and

$$T = \operatorname{diag}\big(\sqrt{a}, \sqrt{b_1/\eta_1}, \ldots\big), \qquad u = \big(\sqrt{a}, \sqrt{b_1\eta_1}, \ldots\big)^\top.$$

The price-first transition becomes the symmetric matrix $J' = TJT^{-1} = \operatorname{diag}(0, r_1, \ldots) + uu^\top$, the input transforms to κu and $J'e_x = \sqrt{a}\,u$, so $h_j = (\kappa/a)e_x^\top (J')^{j+1} e_x$. The spectral decomposition of J' gives nonnegative weights summing to $h_0 = \kappa$, and Schur stability puts the eigenvalues in $[0,1)$. The case $a = 0$ follows by continuity. A nonnegative mixture of decaying geometric sequences is completely monotone. Since J is nonnegative with row sums at most one, every price row of J^t has ℓ_1, hence ℓ_2, norm at most one, attained at $t = 0$.

Theorem 3.1. With zero initial states, $y_{i,t} = \sum_{s\le t} \eta_i r_i^{t-s} x_s$, so $\sum_i q_i y_{i,t} = \sum_{s\le t} k_{t-s} x_s$ and every response is determined by (a, κ, k), hence by M. Conversely, equal responses for all probes imply equal H, hence equal G, and partial fractions of G identify the distinct poles with nonzero residues w_i, i.e. M. After cancellation, $H = \kappa D(z)/[(z-a)D(z) - \kappa N(z)]$ with $D = \prod(z - r_i)$; at a retained r_i the denom-

inator equals $-\kappa w_i r_i \prod_{j\neq i}(r_i - r_j) \neq 0$, so no factor cancels and the minimal degree is $1 + |\mathrm{supp}(M)|$. For (ii), direct algebra gives

$$G_\delta(z) - G_0(z) = -\frac{qz(z-1)\delta^2}{(z-1+\bar{\eta})\big((z-1+\bar{\eta})^2 - \delta^2\big)}.$$

On $|z| = 1$ each factor has modulus at least ϵ. Uniform stability gives $M_H = \sup|H_\theta| < \infty$, and the identity $H_\delta - H_0 = H_\delta H_0(G_\delta - G_0)$ gives an ℓ_2 gain at most $K\delta^2$. The mean difference of a power-constrained probe has squared norm at most $K^2\delta^4 PT$; the equal-noise KL is at most $K^2\delta^4 PT/(2\sigma_\nu^2)$, and Pinsker's inequality with total variation at least $1 - 2\alpha$ gives (8).

Theorem 4.1. In either positive economy $J \geq 0$ and $\|J\|_\infty \leq 1 - \epsilon(1-S) < 1$, so the response is nonnegative with $\max h = \kappa$ and $\sum h = \kappa/(1-S)$. From the display above, $|G_\delta - G_0| \leq 2q\delta^2/\epsilon^3$ on the circle; combined with the resolvent identity and positivity this gives $\|H_\delta - H_0\|_\infty \leq K_*\delta^2$, and Parseval bounds $\|h^\delta - h^0\|_2$ by the same constant. Young's inequality gives $\|\mu_\delta - \mu_0\|_2 \leq K_*\delta^2\|v\|_1$, and Gaussian KL with Pinsker gives (9). For (ii), the mean difference is a lower-triangular convolution matrix times v; each column is a shifted copy of a prefix of d_T, so $\|\mu_\delta - \mu_0\|_2 \leq V\|d_T\|_2$ with equality for a pulse at $t = 0$. For two known equal-covariance Gaussian laws the minimax error is $\Phi(-\|\mu_\delta - \mu_0\|_2/(2\sigma))$, attained by the midpoint likelihood-ratio test. For (iii), the merged measures have equal mass and their first moments differ by $-q\delta^2$, which enters h_2 with factor κ^2 while h_0, h_1 coincide; a pulse A separates the third-observation means by $A\kappa^2 q\delta^2$. Part (iv) follows from the ℓ_2 gain bound, and (v) from superposition and $\max_j h_j = \kappa$, which give $\sup_t|\lambda x_t| \leq |\lambda|\kappa V$, attained by a single pulse.

Lemma 4.2. After the uncontrolled update, the pre-probe displacement at step j is $\sum_{i<j} h_{j-i}v_i$; average linear-book execution adds $\kappa v_j/2$. Summing signed execution costs gives $\frac{1}{2}v^\top T_h v$. Each kernel $\lambda^{|i-j|}$ is a positive-definite AR(1) covariance for $0 \le \lambda < 1$, and a nonnegative mixture with positive total weight is positive definite. For a round trip the fundamental cash term cancels.

Theorem 4.3. Let $U_t = \sum_{j<t} v_j$. Positive responses give $0 \le x_t \le \kappa U_t$ and $I_t = -U_t$. Summing cash and marking inventory, $B - E_t = U_t x_t - \sum_{j<t} v_j x^0_{j+1} - \frac{\kappa}{2}\sum_{j<t} v_j^2 + cU_t$. Its quadratic part $v^\top A_t v$ has diagonal $h_{t-1-i} - \kappa/2$ and off-diagonal $\frac{1}{2}(h_{t-1-i} + h_{t-1-j} - h_{j-i})$ for $i < j$; monotonicity of h bounds every entry by $\kappa/2$. Hence $B - E_t \le \kappa U_t^2/2 + cU_t \le \kappa V^2/2 + cV$, while position value is at most $V(P_* + \kappa V)$; the stated inequality guarantees margin. A single pulse attains both adverse bounds at its first response, so no larger allowance is robust. Solving the quadratic gives (13). The quantity requirement $V_B \gtrsim \delta^{-2}$ then forces $B \gtrsim \delta^{-4}$, and the pulse A funded with $(\mu + \frac{1}{2})\kappa A^2 + (c + \mu P_*)A$ attains it.

Proposition 4.4. The initial cash receipt is common to both alternatives and the dealer then holds $-A$. At time three $E_{3,r} = B + \kappa A^2/2 - cA - A^2 h_{r,2}$, so $E_{3,\delta} - E_{3,0} = A^2\kappa^2 q\delta^2 > 0$.

Theorem 4.5. Marking residual inventory at P_* gives $E_\infty = B - \frac{1}{2}v^\top T_h v - c\sum |v_t| \ge 0$. For each mode $\lambda \le r$ the Toeplitz kernel has spectral density $(1-\lambda^2)/(1-2\lambda\cos\omega + \lambda^2) \ge (1-r)/(1+r)$, so $v^\top T_h v \ge m\|v\|_2^2$ and $\|v\|_2^2 \le 2B/m$. The transfer bound gives mean-difference squared norm at most $K_*^2\delta^4\|v\|_2^2$, hence the KL bound; Pinsker gives (14). For one pulse, cash becomes $B_A - A(P_* + \kappa A/2) - cA$ and inventory A; at later marks $E_t = \mu A P_* + A^2 h_{t-1}$ and $\mu|I_t|P_t = \mu A P_* + \mu A^2 h_{t-1}$, so margin holds for $\mu \le 1$.

Proposition 4.6. In each block $x_1 = w_1$, $x_2 = b_1 w_1 + w_2$ and $x_3 = b_{2,r} w_1 + b_1 w_2 + w_3$, so $|x_t| \le 3b$. Observation noise changes variances but not the covariance $s_w^2 b_{2,r}$. The product variance is the product of marginal variances plus the squared covariance plus a fourth-cumulant term $b_{2,r}^2(\mathbb{E}w^4 - 3s_w^4) \le 0$ for uniform innovations, so it is at most R. Chebyshev bounds each midpoint error by $4R/(n\Delta^2)$.

Proposition 4.7. The CARA certainty equivalent is $s - c_\theta$, whose infimum is $s - \sup_\theta c_\theta$; robust break-even gives the quotes. Integrating $v(V) - c$ between $V_0 - b(s_0 + \Delta)$ and $V_0 - bs_0$ gives the surplus formula.

Proposition 5.1. Parts (i) and (ii) follow from (15) with $\Delta h = \Theta(\delta)$ for a rate shift and $\Theta(\delta^2)$ for symmetric splitting, and from additivity of KL over independent pulse cycles. Part (iii) solves the session constraint for δ; substituting $d_T(\delta) \approx \delta^2 \tilde{d}_T$ gives the fourth-root approximation.

Theorems 5.2 and 5.3. For a flat round trip, telescoping the cash account gives $L_i(v) = \frac{1}{2} v^\top T_{h_i} v = \frac{1}{2\pi} \int g_i(\omega) |V(e^{i\omega})|^2 d\omega$, with $g_i > 0$ by Lemma 2.4. Parseval gives $\|(H_1 - H_0)v\|_2^2 = \frac{1}{2\pi} \int |\Delta\hat{H}|^2 |V|^2 \le \sup_\omega (|\Delta\hat{H}|^2 / g_i) L_i(v)$, and the Neyman–Pearson requirement $\|(H_1 - H_0)v\|_2 \ge 2\sigma z_{1-\alpha}$ gives (19). For adaptive policies, Doob's optional stopping theorem applied to the log-likelihood-ratio martingale and the data-processing inequality give $\mathbb{E}_0[\Lambda_\tau] \ge \mathrm{kl}(\alpha, 1-\beta)$, while $\mathbb{E}_0[\Lambda_\tau] = \mathbb{E}_0 \sum_t (\Delta\mu_t)^2/(2\sigma^2) \le \sup(|\Delta\hat{H}|^2/g_0)\mathbb{E}_0[L_0]/(2\sigma^2)$, which gives (20).

Theorem 5.4. Write a feasible flat probe as $v = Eu$ and decompose $u = r\, d(\theta) + u_\perp$ with $d(\theta) = \cos\theta\, e_{L-1} + \sin\theta\, e_L$ in the G_i-orthonormal eigenbasis. With $\lambda(\theta) = \lambda_{L-1} \cos^2\theta + \lambda_L \sin^2\theta$ and $L_0(\theta) = c_{\mathrm{req}}/\lambda(\theta)$, the excess loss $\Delta_2 = u^\top G_i u - L_0(\theta)$ satisfies $\Delta_2 \ge \sum_{j \le L-2} (1 - \lambda_j/\lambda(\theta)) c_j^2 \ge 0$, so the residual is controlled by Δ_2. Hölder's

inequality with a dual sign vector s gives $c_{\text{fee}}\|Eu\|_1 \geq c_{\text{fee}} s^\top E(rd(\theta) + u_\perp)$, and Cauchy–Schwarz on the residual term gives a lower bound that is concave in $\sqrt{\Delta_2}$. Minimizing the quadratic in $\sqrt{\Delta_2}$ in closed form and replacing the θ partition by the smallest singular value of $[Ee_{L-1}, Ee_L]$ gives (21). The certified numbers in the text come from an outward-rounded interval evaluation over 1,800 θ-intervals and 4,000 Δ_2-intervals with a Weyl enclosure of the eigenvalue error.

Corollary 5.5. Since $\Delta h_t = O(\delta^2)$, the variance difference $\sigma_{1,t}^2 - \sigma_{0,t}^2 = \sigma^2 \xi[(h_1 + h_0) * v]_t [\Delta h * v]_t = O(\delta^2)$, and the variance term of the Gaussian KL, $\frac{1}{2}\sum(x - 1 - \log x)$ with $x = \sigma_{0,t}^2/\sigma_{1,t}^2$, is quadratic in that difference. Convex impact only increases dissipation.

Proposition 6.1. Standard Cramér–Rao for a Gaussian mean model with Jacobian J. For $K = 2$ the columns of J for the two rates become collinear as $\Delta\eta \to 0$; expanding to third order, the smallest singular value of J is of order $(\Delta\eta)^3$, so $[F^{-1}]_{kk} = \Theta((\Delta\eta)^{-6})$.

Theorem 6.2. Expanding $M_m(\theta_1) - M_m(\theta_0)$ under (23) gives zero for $m \leq 2$ and $4\tau p_- p_+ (\Delta\eta)^3$ for $m = 3$. The map from moments to impulse responses is analytic with nonsingular derivative at the common limit, so $\|\Delta h\|_2 = \Theta(\tau(\Delta\eta)^3)$. Le Cam's two-point lemma with N independent Gaussian experiments gives (25). For (iii), the projection estimator of M_3 is unbiased with variance $\sigma_h^2/(N\|\nabla M_3 h\|^2)$, and sub-Gaussian concentration gives the sample size; the Prony–Hankel determinant identity follows from $g_m = w_1\eta_1 r_1^m + w_2\eta_2 r_2^m$.

Theorem 6.3. Let (s^A, ω^A) and (s^B, ω^B) be the K-point Gauss and Gauss–Radau rules for the uniform weight on $[-1, 1]$, normalized to unit mass. Both are positive

and exact for polynomials of degree $2K-2$, and their moments of degree $2K-1$ differ by $g_K \neq 0$. With $r = r_c + \Delta\eta\, s$ and weights $W\omega$, the raw moments $\sum w_i r_i^m$ are polynomials in $\Delta\eta$ that agree below degree $2K-1$, so the kernels differ by $O((\Delta\eta)^{2K-1})$ with nonzero leading coefficient; analyticity of the kernel-to-response map gives $\|h^A - h^B\|_2 \asymp (\Delta\eta)^{2K-1}$, and Le Cam's lemma gives the bound.

Lemma 6.5. Let $L = 2n-2$ and let p^* be the best uniform approximation of $|x|$ on $[-1,1]$ by polynomials of degree L, with error $E = E_L(|x|)$. By Chebyshev's equioscillation theorem (Rivlin, 1981) there are points $t_0 < \cdots < t_{L+1}$ at which $|x| - p^*(x)$ equals $\pm E$ with alternating signs. The divided-difference weights $w_i = 1/\prod_{j\neq i}(t_i - t_j)$ alternate in sign and satisfy $\sum_i w_i p(t_i) = 0$ for every polynomial p of degree at most L; multiplying by -1 if necessary, w_i has the sign of $|t_i| - p^*(t_i)$ for every i. Exactly n weights are positive and n negative, and the positive and negative parts have equal mass because constants are annihilated. Normalizing each part to unit mass and mapping $x \mapsto \rho(x+1)/2$ gives n-atom probability measures π_A (positive part) and π_B (negative part) on $[0,\rho]$ with equal moments of orders $0,\dots,L$. For the 1-Lipschitz function $f(r) = |r - \rho/2| = (\rho/2)|x|$,

$$\int f\, d(\pi_A - \pi_B) = \frac{\rho}{2}\int \big(|x| - p^*(x)\big)\, d(\pi_A - \pi_B) = \frac{\rho}{2}(E + E) = \rho E,$$

because p^* integrates to the same value under both measures, $|x| - p^* = E$ on the support of π_A and $-E$ on that of π_B. By Kantorovich–Rubinstein duality $W_1(\pi_A, \pi_B) \geq \rho E \geq c_*\rho/L$.

Theorem 6.6. Write $K(z) = \sum_m k_m z^m$, $D = 1 - az - \kappa z K$ and $H = \kappa/D$, which is the generating function of the recursion $h_t = a h_{t-1} + \kappa \sum_m k_m h_{t-1-m}$ with $h_0 = \kappa$. Since $k_m \geq 0$ and $\sum_m k_m = \int (1-r)^{-1} d\mu \leq W/(1-\rho)$, the ℓ_1 norm of $az + \kappa z K$ is

at most $s < 1$, so $\|H\|_1 \le \kappa/(1-s)$ and $\|1/H\|_1 = \|D\|_1/\kappa \le (1+s)/\kappa$.

(i) Let $\mu_A = W\pi_A$ and $\mu_B = W\pi_B$ with π_A, π_B from Lemma 6.5; both lie in $\mathcal{M}(\rho, W)$. With $\nu = \mu_A - \mu_B$, $\sum_m (k_m^A - k_m^B) z^m = \int (1-rz)^{-1} d\nu(r) = z^{2n-1} \int r^{2n-1}(1-rz)^{-1} d\nu(r)$, because the first $2n-1$ terms of the geometric series integrate to zero. On $|z| = 1$ the modulus is at most $2W\rho^{2n-1}/(1-\rho)$, so Parseval gives $\|k^A - k^B\|_2 \le 2W\rho^{2n-1}/(1-\rho)$. Next, $H_A - H_B = z(K_A - K_B)H_A H_B$, so by Young's inequality $\|h^A - h^B\|_2 \le \|H_A\|_1 \|H_B\|_1 \|k^A - k^B\|_2 \le C_h \|k^A - k^B\|_2$. The Kullback–Leibler divergence between the N-sample Gaussian experiments is $N\|h^A - h^B\|_2^2/(2\sigma^2) \le (N/\sigma^2)\, 2C_h^2 W^2 \rho^{2(2n-1)}/(1-\rho)^2$, which is at most one as soon as $2(2n-1)\log(1/\rho) \ge \Lambda_N - 2\log(1/\rho)$. Take n the smallest integer with this property and $n \ge 2$; then $2n - 2 < \Lambda_N/(2\log(1/\rho))$ when $\Lambda_N \ge 4\log(1/\rho)$. Since d is a norm on signed measures and $d(\mu_A, \mu_B) = W\, W_1(\pi_A, \pi_B) \ge W c_* \rho/(2n-2) > 2c_* \rho W \log(1/\rho)/\Lambda_N$, the Bretagnolle–Huber form of Le Cam's lemma (Tsybakov, 2009, Lemma 2.6 and Theorem 2.2) gives $\max_{j \in \{A,B\}} P_j(d(\hat{\mu}, \mu_j) \ge c_* \rho W \log(1/\rho)/\Lambda_N) \ge \frac{1}{4} e^{-1}$.

(ii) *Approximation.* Let f be 1-Lipschitz on $[0, \rho]$ with $f(0) = 0$, so $|f| \le \rho$. The function $g(x) = f(\rho(x+1)/2)$ is $(\rho/2)$-Lipschitz on $[-1, 1]$, and Jackson's theorem (Rivlin, 1981) gives a polynomial p of degree L with $\sup|g - p| \le (\pi/2)(\rho/2)/(L+1) \le \rho$, hence $\sup|p| \le 2\rho$. Its Chebyshev coefficients satisfy $|b_j| \le 2\sup|p| \le 4\rho$. The polynomial $q(r) = p(2r/\rho - 1)$ approximates f with the same error, and its monomial coefficients satisfy $\sum_j |q_j| \le \sum_{j \le L} |b_j|\, \|T_j(2r/\rho - 1)\|_{\mathrm{coef}} \le 4\rho \sum_{j \le L} B(\rho)^j \le \rho B(\rho)^{L+1}$, where $\|\cdot\|_{\mathrm{coef}}$ is the sum of absolute monomial coefficients, $\|T_j\|_{\mathrm{coef}} \le (1+\sqrt{2})^j$, substituting $y = \alpha r - 1$ multiplies the coefficient sum of y^i by at most $(1+\alpha)^i$, and $B(\rho) - 1 \ge 4$. *Moment comparison.* If $\max_{j \le L} |\int r^j d\hat{\mu} - k_j| \le 2\delta$, then

$\hat{\mu}$ has mass at most $W + 2\delta$ and

$$\int f\, d(\hat{\mu} - \mu) = \int (f - q)\, d(\hat{\mu} - \mu) + \sum_{j \le L} q_j \Big(\int r^j d\hat{\mu} - k_j \Big) \le \frac{\pi\rho(W + \delta)}{2(L+1)} + 2\delta\rho B(\rho)^{L+1}.$$

Kernel recovery. From $\kappa z K = 1 - az - \kappa/H$, the kernel values $k_0, \dots, k_L$ are the coefficients of $(1 - az - \kappa/H)/(\kappa z)$, which depend on $h_0, \dots, h_{L+1}$ only. Work in the algebra of power series modulo z^{L+2} with the ℓ_1 norm, which is submultiplicative. If the averaged response is $\hat{H} = H + E$ and $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$, then $1/\hat{H} = (1/H)(1 + E/H)^{-1}$ and $\|1/\hat{H} - 1/H\|_1 \le 2\|1/H\|_1^2 \|E\|_1$, so $\max_j |\hat{k}_j - k_j| \le \delta \equiv 2((1+s)/\kappa)^2 (L+2) \max_{t \le L+1} |E_t|$. Each E_t is Gaussian with variance σ^2/N, so with probability at least $1 - \beta$, $\max_t |E_t| \le \sigma \sqrt{2 \log(2(L+2)/\beta)/N}$. The true μ is feasible for the linear program with value at most δ, so the minimizer satisfies the moment condition above. *Choice of L.* With $L + 1 = \lfloor \log(N/\sigma^2)/(4 \log B(\rho)) \rfloor$, $B(\rho)^{L+1} \le (N/\sigma^2)^{1/4}$, so $\delta B(\rho)^{L+1} = O((\sigma^2/N)^{1/4} L \sqrt{\log(L/\beta)})$ and $\delta/(L+1)$ are $o(1/\log(N/\sigma^2))$, while $\pi\rho W/(2(L+1)) \le 2\pi\rho W \log B(\rho)/(\log(N/\sigma^2) - 4 \log B(\rho))$. For N/σ^2 larger than a threshold depending on $(\rho, W, a, \kappa, \beta)$ the side condition $\|1/H\|_1 \|E\|_1 \le \frac{1}{2}$ holds on the event and the total is below $3\pi\rho W \log B(\rho)/\log(N/\sigma^2)$, uniformly over $\mathcal{M}(\rho, W)$. The final statement combines (i) and (ii).

Theorem 6.8. For stationary Gaussian processes with spectral densities $f_j \propto |H_j(e^{i\omega})|^2$, the divergence rate is

$$D = \frac{1}{4\pi} \int_{-\pi}^{\pi} \big(\varrho - 1 - \log \varrho \big)\, d\omega, \qquad \varrho = f_1/f_0 = |H_1/H_0|^2.$$

By the resolvent identity, $H_1/H_0 - 1 = H_1 \Delta G$ with $\Delta G = G_1 - G_0$, and uniform stability bounds $|H_1|$ on the circle, so $u \equiv \varrho - 1 = 2\,\mathrm{Re}(H_1 \Delta G) + O(|\Delta G|^2)$. Write $G(z) = \sum_k w_k \phi(z, \eta_k)$, with w_k the cohort weights (any factor of η_k is absorbed in ϕ),

where ϕ is analytic in η on a neighbourhood of $[\eta_{\min}, \eta_{\max}]$ uniformly for $|z| = 1$ (its pole $z = 1-\eta$ stays inside the disk). Expanding ϕ around $\bar{\eta}$, $\Delta G = \sum_{j\geq 0} \frac{\partial_\eta^j \phi(z,\bar{\eta})}{j!} \Delta\tilde{M}_j$, where $\tilde{M}_j = \sum_k w_k(\eta_k - \bar{\eta})^j$. For the pair, $\Delta\tilde{M}_0 = \Delta\tilde{M}_1 = 0$ and $\Delta\tilde{M}_2 = q\delta^2$; for the moment-matched economies, $\Delta\tilde{M}_j = 0$ for $j \leq 2$ and $\Delta\tilde{M}_3 = 4\tau p_- p_+ (\Delta\eta)^3$ by (24); higher terms are smaller by a factor of the spacing. Hence $\sup_\omega |u| = O(\delta^2)$, respectively $O(\tau(\Delta\eta)^3)$, and $u = c(\omega)\epsilon + o(\epsilon)$ with $\epsilon = \delta^2$ or $\tau(\Delta\eta)^3$ and $c(\omega) = 2\,\mathrm{Re}\big(H_0\,\partial_\eta^j \phi(\cdot,\bar{\eta})\big)/j!$, $j = 2$ or 3. The function c does not vanish identically: $H_0\partial_\eta^j\phi$ is analytic outside the disk and non-constant, and an analytic function whose real part vanishes on the circle is an imaginary constant. For $|u| \leq \frac{1}{2}$, $u^2/4 \leq u - \log(1+u) \leq u^2$, so $D \asymp \int c(\omega)^2 d\omega\, \epsilon^2$, which gives (i) and (ii). The sample requirements follow from $D(P_1^T \| P_0^T) = TD + O(1)$ for stationary Gaussian processes with smooth spectra and the Bretagnolle–Huber inequality (Tsybakov, 2009, Lemma 2.6); attainment follows from Stein's lemma for the likelihood-ratio test.

Theorem 6.7. The zeros of $H = \kappa D/[(z-a)D - \kappa N]$ are the roots of D, which do not cancel; stability puts the poles inside the disk. A rational spectral density $\sigma^2|H(e^{i\omega})|^2$ has a unique minimum-phase factor up to a unimodular constant, which is H; Theorem 3.1 then gives the merged measure.

Proposition 7.1. With tagged flow, $u_{i,t} = r_i u_{i,t-1} + w_i x_t$ involves only cohort i. With measurement noise in u, instrumenting $u_{i,t-1}$ by $u_{i,t-2}$ gives a $\sqrt{T}$-consistent estimator whose asymptotic variance depends on (r_i, w_i) and the price process but not on $|r_i - r_j|$.

Theorem 8.1. With J_θ upper triangular the eigenvalues are fixed. The stationary covariance solves $P = J_\theta P J_\theta^\top + \mathrm{diag}(s_1, s_2)$, and P_{11} is affine in s_2 with positive slope, so for each θ a unique $s_2(\theta) \geq 0$ gives $P_{11} = V$ when V exceeds the value at $s_2 = 0$.

The one-step price variance is s_1 throughout, and $e_1^\top J_\theta e_2 = \theta$ gives $\sup_k \|e_1^\top J_\theta^k\| \geq |\theta|$.

Proposition 9.1. $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, so $\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)(p_b - m_b) = (1-\alpha)x_b^{\mathrm{EMA}}$.

B Notation

Symbol	Meaning
x_t	pricing error (price minus fundamental)
a	price persistence (anchor), $0 \leq a < 1$
κ	price impact per unit of executed quantity
$q_i,\ \eta_i,\ r_i$	cohort weight, adaptation rate, memory rate $r_i = 1 - \eta_i$
$w_i = q_i\eta_i$	merged weight
$M = \sum_i w_i \delta_{r_i}$	merged rate–weight measure
$k_m,\ G(z)$	feedback kernel and its transform
$H(z),\ h_j$	closed-loop transfer function and impulse response, $h_0 = \kappa$
$J,\ S$	state transition matrix and $S = a + \kappa \sum q_i$
$v_t,\ V$	executed probe quantity and total quantity $\sum \lvert v_t \rvert$
d_T	impulse-response difference between hypotheses, first T taps
σ	observation-noise standard deviation
$\delta,\ \Delta\eta$	half-separation of a known pair; spacing of unknown cohorts
K_*	transfer-difference constant $2\kappa^2 q/[\epsilon^3(1-S)^2]$
T_h	Toeplitz matrix $(h_{\lvert i-j \rvert})$ of execution cost
$g_i(\omega)$	dissipation kernel $\mathrm{Re}\,\hat{H}_i(e^{i\omega}) - \kappa/2$

Symbol	Meaning
$B,\ \mu,\ c,\ \phi$	capital, maintenance-margin fraction, proportional cost, taker fee
V_{eff}	per-trade execution capacity $\min(V_{\max}, D_{\max}/\kappa)$
$\hat{Q},\ \hat{S}$	reduced-form loop gain and persistence indices

Supplemental Appendix for "Auditable Memory and Financial Identification Limits"

C Internet appendix

C.1 A finite-window price-surveillance bound

Theorem C.1 (Finite-window abstention). *Observe* $x_t = sx_{t-1} + w_t$, $w_t \sim N(0, \sigma^2)$, $x_0 = 0$, $t \le T$, *with* $s \in (0, 2)$. *Let a possibly randomized rule* Ψ *output Stable, Unstable or Abstain, with* $\sup_{s<1} P_s(\Psi = \text{Unstable}) \le \alpha$ *and* $\sup_{s\ge 1} P_s(\Psi = \text{Stable}) \le \alpha$, $\alpha < 1/2$. *Choose* $0 < c_0 < 1$ *with* $c_0 e^{c_0}/(2\sqrt{2}) < 1 - 2\alpha$. *On* $B_T = (1 - c_0/T, 1 + c_0/T)$,

$$\inf_{s\in B_T} P_s(\Psi = \text{Abstain}) \;\ge\; 1 - 2\alpha - \frac{c_0 e^{c_0}}{2\sqrt{2}} > 0,$$

and for every finite T *the minimax worst-case error of binary rules over* $s \in (0, 2)$ *is at least* $1/2$.

Proof. Pair $s_0 \in (1 - c_0/T, 1)$ with $s_1 = s_0 + c_0/T$. Then $D_{\mathrm{KL}}(P_{s_0} \| P_{s_1}) = \frac{(s_1 - s_0)^2}{2\sigma^2} \sum_t \mathbb{E} x_{t-1}^2 \le c_0^2 e^{2c_0}/4$, so Pinsker bounds total variation by $c_0 e^{c_0}/(2\sqrt{2})$. Error control gives $P_{s_0}(\text{Stable}) \le \alpha + \text{TV}$ and $P_{s_0}(\text{Unstable}) \le \alpha$, which yields the abstention bound. For binary rules, continuity in total variation and Le Cam's two-point bound give worst-case error tending to $1/2$. □

The interval B_T is an abstention layer that any error-controlled monitor must respect in the scalar Gaussian case; richer disclosure can remove it.

C.2 Blind inversion of the closed loop

1. *Impulse decoupling:* $g_t = h_{t+1} - ah_t$.

2. *Minimal realization:* form the Hankel matrix $H_{ij} = g_{i+j-1}$ and run Kung's eigensystem realization with a truncated SVD.

3. *Transmission zeros:* compute the zeros of the realization, which under known (κ, a) isolate the roots of the denominator of G.

4. *Weights:* with recovered rates, project $u_t = (x_t - a x_{t-1})/\kappa$ on the reconstructed filters $y_{k,t-1}$.

Without noise each step is exact: the realization has the order of the true system, its transmission zeros are the memory rates (Theorem 3.1), and the projection returns the weights, so the procedure removes the pole-shift bias of fitting poles directly to the price response. With noise, per-cohort recovery degrades as rates move together at the rate given by Theorems 6.3 and 6.4, which is why we do not report it as an estimator of individual cohorts.

C.3 Probing protocols

C.4 Coordinate-weighted output

For $J = \left(\begin{smallmatrix} a & \kappa q \\ \eta a & 1-\eta+\eta\kappa q \end{smallmatrix}\right)$ with $a = 0.5$, $\kappa = 0.8$, $q = 0.2$, $\eta = 0.3$ ($\rho(J) = 0.8224$, non-normal), the price output has $\sup_t \|e_x^\top J^t\| = 1$. The weighted output $C_w = (1, \lambda/\mathrm{MMR}) = (1, 160)$ with $\lambda = 0.8$ and $\mathrm{MMR} = 0.005$ has $\sup_t \|C_w J^t\| = \|C_w\| = 160.003$, attained at $t = 0$, then 122.32, 98.32, 80.79 [RECOMPUTED IN CLOSED FORM]. The factor 160 is a static coordinate weight; $C_w z$ is not a margin quantity because z contains no inventory. Margin must be computed on a cash-and-inventory ledger.

Table 24: Probing protocols at observation horizon $T = 4$ per cycle

Protocol	δ	Executed vol.	Max trade	Error (SE)	Capital
A: single pulse (unconstrained)	0.03	131.5	131.5	0.047 (0.003)	6,985.91
	0.05	47.3	47.3	0.054 (0.004)	921.73
	0.08	18.5	18.5	0.053 (0.004)	146.73
	0.12	8.2	8.2	0.044 (0.003)	31.45
B: repeated capped pulses	0.03	2,768.8	6.2	0.050 (0.003)	7,035.75
	0.05	362.5	6.2	0.054 (0.004)	923.88
	0.08	56.2	6.2	0.047 (0.003)	146.00
	0.12	12.5	6.2	0.038 (0.003)	34.88
C: sequential test, capped	0.03	1,410.2	6.2	0.045 (0.003)	3,585.14
	0.05	195.6	6.2	0.038 (0.003)	500.00
	0.08	35.9	6.2	0.032 (0.003)	94.30
	0.12	10.5	6.2	0.011 (0.002)	29.85
D: generalized eigenvector	0.03	2,359.7	3.7	0.050 (0.003)	2,632.77
	0.05	314.8	3.7	0.048 (0.003)	353.67
	0.08	49.2	3.7	0.046 (0.003)	57.53
	0.12	9.9	3.8	0.045 (0.003)	13.56

2,000 replications per hypothesis and cell. Log–log capital slopes: −3.90, −3.84, −3.47, −3.81. Protocols A and B reproduce the earlier version exactly. C and D are re-implemented because the earlier code was not retained: C is a Wald test with capped pulses; D repeats the top generalized eigenvector of a flat $(T + 1)$-step probe, scaled to a maximum trade of 3.75, and is cheaper than the earlier D. Kind: [RECOMPUTED] (`recompute_p1_protocols.py`).

D Reproducibility: source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part N" below and in table notes refers to notebook N (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

Result	Source	Kind
Complete monotonicity, 892 economies	`L26_complete_monotonicity`	recomputed
$d_2 = -\kappa^2 q \delta^2$	`P1.7-T1_third_tap`	recomputed
Ledger identity; T_h positive definite	`R2-T1`, `R2-T3`	recomputed
Figure 2	`make_figs2.py`	recomputed
$\delta_{\min}(3)$, Table 4 recomputed columns, 61.14, 29.94	`recompute_theory.py` → `R_P1_theory_constants.json`	recomputed
Pulse benchmarks (271, 58, 12, 11 pulses), Figures 1–3	`make_figs3.py` → `R_P1_pulse_benchmarks.json`	recomputed
Table 2, Table 10, worked example, passive simulation	`recompute_extra.py` → `R_P1_extra.json`	recomputed
Daily noise (Table 7)	`P1.3-T1_sigma_obs`	real data
Table 9 recomputed columns	same	recomputed
K-cohort slopes (Table 11)	`T1_K_cohort_rate`	recomputed
ESPRIT slope −6.15	`P1.2-T1_esprit_rate`	recomputed
Minimum phase, 1,733 economies; rank 6 of 6	`T2a`, `T2b`	recomputed
Passive slopes −4.21, −5.93	`T3_passive_rate`	recomputed
Tagged-flow slope 0.011	`T4_disclosure`	recomputed
Spectral blindness gains	`T5_spectral_blindness`	recomputed
Noise and capital grid (Table 6)	`P1.3-T1_sigma_obs`	real data
Loop gain and placebo (Table 20)	`P1.6-T1_ema_placebo`	real data
Estimator table (Table 21)	`P1.5-T1`, `T2`, `T3`	recomputed

Result	Source	Kind
Data integrity (111 files)	`R2-T9_data_integrity`	audit
Tables 3, 4 (certified, feasible, adaptive columns)	`recompute_p1_capital.py` → `R_P1_capital_ledger.json`	recomputed
Table 14; resolution floors; Table 24	`recompute_p1_protocols.py` → `R_P1_protocols.json`	recomputed
SPRT capital figures	`recompute_p1_sprt.py` → `R_P1_sprt.json`	recomputed
Table 8 (N^*, σ_{crit})	`recompute_crb.py` → `R_P1_crb.json`	recomputed
Table 8 (ML columns), Table 9 (estimators)	`recompute_p1_estimators.py`	recomputed
Symmetric multi-asset network (Section 8)	`recompute_oct20.py`	recomputed
Proof audit: K_* bound and d_2 identity (Theorem 4.1), moment gaps (Theorem 6.2), spectral-blindness example (Theorem 8.1)	`recompute_audit.py` → `R_audit_oct2026.json`	recomputed
Theorem 6.4, Table 12	`kcohort_upper.py` → `R_P1_kcohort_upper.json`	recomputed
Lem. 6.5, Thm 6.6, Tab. 13	`general_minimax.py`	recomputed
Theorem 7.2 numbers	`disclosure_value.py` → `R_P1_disclosure_value.json`	recomputed
Tables 18, 19, 22	Part 8: `V2_spot_panel`, `V3_fee_events`	real data

Result	Source	Kind
Table 15, top panel	Part 4: `X9_disclosure_metrics`	real data
Table 15, bottom panel; class-position forecasts	Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…)	real data
Table 16	Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…)	real data
Basis-trade signature; real-data Cochrane–Orcutt	Part 13: `R6`, `R7`	real data (post hoc)
Feedback across 30 perpetuals	Part 4: `X3_T6_cross_section`	real data